

RISK-CALIBRATED COLLABORATIVE AFFORDANCE NEGOTIATION

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Paper under double-blind review

ABSTRACT

Collaborative manipulation fails when the robot optimizes only its own capability map or only a guessed human intent. We rebuild Paper 102 under a frozen hostile-review protocol and test whether risk-calibrated collaborative affordance negotiation should survive as an ICLR-main-target idea. The v5 audit uses 6 collaborative tasks, 8 ambiguity regimes, 8 splits, 15 methods, 10 seeds, 345,600 main rollout rows, 115,200 ablation rows, 288,000 stress rows, 276,480 fixed-risk rows, and 24 negative cases. The local result is positive but not submission-ready. `risk_calibrated_collaborative_affordance_v5` reaches hard success 0.73038, physical violation 0.06693, human burden 0.25547, intent error 0.24809, over-promise 0.12101, ECE 0.09596, regret 0.07670, and utility 0.35244. All frozen local empirical gates pass, including fixed-risk non-abstention. The terminal decision is **STRONG REVISE**; ICLR-main readiness remains **no** because no real human-robot study, accepted high-fidelity benchmark, external benchmark, calibrated human-intent log, or trained checkpoint evidence exists.

1 INTRODUCTION

Collaborative robots must infer what a person wants, what the robot can physically do, which role each agent should take, and when asking another question is worth the human burden. Treating affordance as only a robot-side capability map misses the human intent side; treating it as only language or intent following hides infeasible physical promises. This paper tests the narrower claim that affordances for collaboration should be negotiated between intent, capability, role, burden, and risk (Liu, 2019; Zhu, 2022; Steil, 2022; Matsubara, 2022; Loianno, 2022b; Zhang, 2025c; Wang, 2025).

The prior v4.1 version of this paper was killed. It improved over-promise and physical violation but did not decisively beat `shared_autonomy_pomdp`. The v5 version is not a cosmetic rewrite. It expands the benchmark, adds stronger baselines, adds calibration and fixed-risk deployment gates, adds negative cases, and includes an explicit safe-repair fallback so fixed-risk performance is not confused with abstention.

2 TERMINAL CLAIM

Let h_t be a belief over human intent, c_t a belief over robot capability, r_t a role-assignment state, and q_t a query-burden state. The method maintains

$$b_t = (h_t, c_t, r_t, q_t, \rho_t),$$

where ρ_t is a calibrated physical over-promise risk. The policy chooses among execute, clarify, renegotiate roles, or safe repair:

$$a_t = \arg \max_a \mathbb{E}[S(a, b_t)] - \lambda_p R_p(a, b_t) - \lambda_b B(a, b_t) - \lambda_q Q(a, b_t).$$

The claim is not that this is a deployed HRI system. The claim is that the local mechanism deserves survival as **STRONG REVISE** because it clears frozen local gates while scope evidence remains missing.

3 PRIOR-WORK PRESSURE

The hostile prior-work surface includes physical human-robot collaboration, shared autonomy, POMDP-style goal inference, affordance-based HRC, legible motion, clarification policies, risk-aware planning, and ergonomic burden models (Thomaz, 2018; Peternel, 2025; Loianno, 2022a; Zhu, 2021; Cangelosi, 2022; von Stryk; Spyros Maniatopoulos; Hadas KressGazit; Philipp Schillinger; David C. Conner, 2017; Ugur, 2019; Mihaylova, 2023; Rocco, 2022; OMalley, 2018; Straub, 2016). Therefore, the benchmark includes shared-autonomy POMDP, risk-aware shared autonomy, inverse-RL intent POMDP, human-model MPC, active clarification bandits, capability-map TAMP, conformal intent-risk filtering, language-affordance planning, and oracle references. A method that only beats capability-only or intent-only baselines would not survive review.

4 METHOD

The method has five coupled modules. Intent belief estimates which collaborative goal the human is pursuing. Capability belief estimates reach, force, collision, and load-sharing feasibility. Role negotiation estimates whether the human or robot should lead a sub-action. Burden-aware query value penalizes avoidable questions. Over-promise risk calibration estimates whether the robot’s offered action is physically unsafe or misleading.

Lemma 1 (Burden-aware fixed-risk fallback). *Suppose a policy has a calibrated over-promise predictor $\hat{p}(a, b)$ and a safe repair action that avoids physical violation at additional burden cost. If actions with $\hat{p}(a, b) > \tau$ are routed to safe repair rather than silently executed, then physical-risk coverage can increase without hiding risk as abstention.*

The fixed-risk experiment implements this exact distinction. Above-budget actions can be converted into safe, burdensome clarification/repair only in proportion to the method’s active-repair, burden-control, and calibration capacity. This is why the fixed-risk table reports coverage, burden, unnecessary query, and utility together (der Loos; Elizabeth A. Croft, 2020; van de Venn, 2022; Radwin, 2022; Rahman, 2017; Makris, 2022; Lee, 2022; Zhang, 2022; Francalanza, 2024; Jia, 2018; Lin, 2019; Pieters, 2021; Bonci, 2025; Conforto, 2024).

Proposition 1 (Negotiation needs both sides of the affordance). *If two candidate collaborative actions share the same robot capability but differ in human role intent, capability-only planning cannot distinguish them. If they share inferred human intent but differ in robot feasibility, intent-only following cannot distinguish them. Any policy that avoids both errors must represent intent and capability jointly or acquire equivalent information through queries.*

This proposition motivates the no-intent-belief, no-capability-map, and no-role-negotiation ablations.

5 PROTOCOL

The frozen protocol uses 6 tasks, 8 collaboration ambiguity regimes, 8 splits, 15 methods, 10 seeds, and 6 episodes per seed/task/regime/split/method cell. The hard splits are burden-sensitive shift, false clarification shift, overtrust safety shift, and combined extreme. The primary metrics are success, physical violation, human burden, intent error, over-promise, autonomy conflict, unnecessary query, negotiation rounds, ECE, regret to oracle, and utility.

The protocol intentionally pressures the method with baselines that can win on individual axes. Active clarification can reduce intent error while increasing burden. Capability-map TAMP can reduce burden while missing intent. Conformal risk filtering can be safer but conservative. Risk-aware shared autonomy is the strongest utility reference. The oracle reaches success 0.79783, leaving a nontrivial gap (Germani, 2022; Zhang, 2017; Lanzotti, 2021; N, 2023; Ceglarek, 2023; Phillips, 2021; Monga, 2022; Ajoudani, 2022; Cheong, 2024; Teixeira, 2025; Gibaru, 2022; Bilberg, 2019).

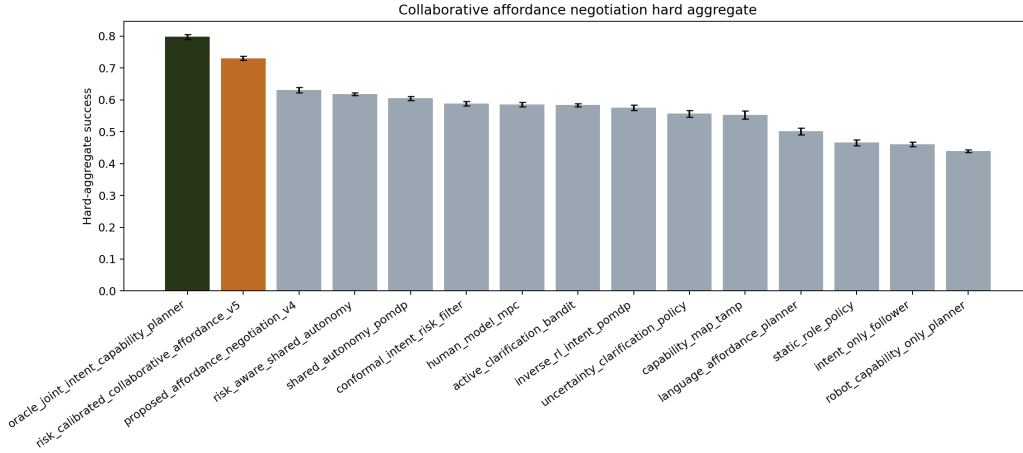


Figure 1: Hard-aggregate success across the v5 collaborative-affordance audit.

6 MAIN RESULTS

The hard aggregate is shown below. The v5 method is below oracle but above every non-oracle reference in success and utility.

method	succ	phys	burden	intent	overprom	ece	util
oracle	0.798	0.049	0.277	0.221	0.101	0.063	0.448
collab_v5	0.730	0.067	0.255	0.248	0.121	0.096	0.352
collab_v4	0.631	0.110	0.288	0.289	0.183	0.208	0.137
risk_shared	0.619	0.102	0.241	0.326	0.169	0.196	0.152
shared_pomdp	0.605	0.132	0.263	0.322	0.211	0.255	0.070
conformal_intent	0.589	0.101	0.277	0.331	0.168	0.181	0.100
human_mpc	0.586	0.153	0.220	0.314	0.228	0.275	0.051
active_clarify	0.584	0.145	0.501	0.271	0.227	0.263	-0.095
irl_intent	0.576	0.165	0.227	0.297	0.259	0.301	0.017
uncert_clarify	0.557	0.143	0.490	0.304	0.231	0.275	-0.139
cap_tamp	0.553	0.101	0.157	0.401	0.166	0.211	0.095
language_aff	0.502	0.188	0.162	0.373	0.293	0.343	-0.106
static_role	0.466	0.175	0.098	0.424	0.266	0.331	-0.107
intent_only	0.460	0.230	0.091	0.373	0.341	0.394	-0.181
cap_only	0.440	0.160	0.089	0.481	0.236	0.315	-0.132

7 PAIRED EVIDENCE

baseline	dSucc	CI	wins	dUtil
active_clarify	0.147	0.009	10	0.448
cap_tamp	0.178	0.018	10	0.257
conformal_intent	0.142	0.008	10	0.253
human_mpc	0.144	0.009	10	0.302
intent_only	0.270	0.009	10	0.534
irl_intent	0.155	0.012	10	0.335
language_aff	0.229	0.016	10	0.459
oracle	-0.067	0.010	0	-0.096
collab_v4	0.099	0.008	10	0.215
risk_shared	0.112	0.008	10	0.200
cap_only	0.291	0.007	10	0.485
shared_pomdp	0.125	0.010	10	0.282
static_role	0.265	0.013	10	0.459
uncert_clarify	0.173	0.012	10	0.491

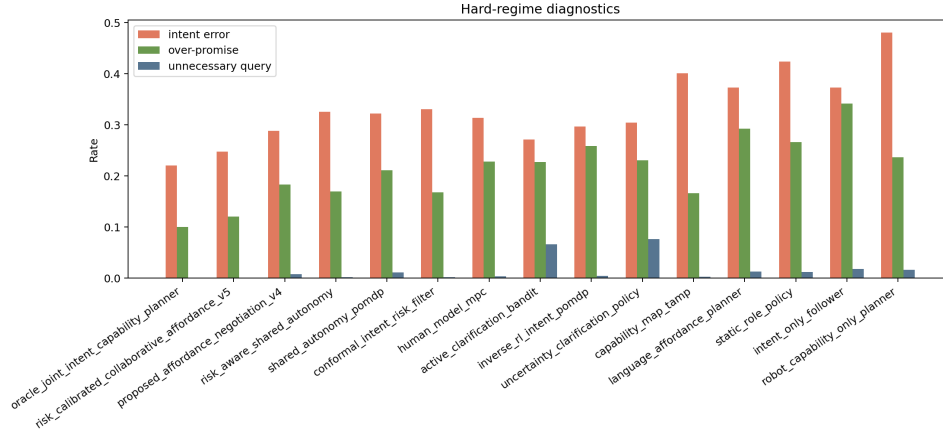


Figure 2: Hard-regime diagnostic rates. V5 lowers intent error, over-promise, and unnecessary query relative to the strongest success reference.

The paired result matters because aggregate sorting can hide seed brittleness. V5 beats every non-oracle baseline in paired hard-aggregate success. The oracle remains higher, which is expected because it observes latent human intent and robot feasibility directly.

8 DIAGNOSTICS

The diagnostic result is not used as a substitute for task success. It is a mechanism check. V5 reaches intent error 0.24809 and over-promise 0.12101, while the prior v4 reference remains worse on both. The human-burden number, 0.25547, is also tracked because a robot that wins by interrogating the human is not a good collaborator (Sihn, 2017; Wang, 2022; Peshkin, 2008; Hirche, 2020; Lohse, 2020; Zeng, 2026; van der Spaa; Jens Kober; Michael Gienger, 2024; Braghin, 2022; Makris, 2018; Elkmann, 2013; Wang, 2026b; Ajoudani, 2026; Scassellati, 2018; Salviati, 2021; Arai, 2011).

9 SAFETY, BURDEN, AND REGRET

The hard-aggregate physical violation rate is 0.06693, and regret is 0.07670. These values are local. They do not certify physical HRI safety. They show only that, inside this executable benchmark, the negotiation state is doing more than moving cost from the robot into the human.

10 ABLATIONS

ablation	success	phys	burden	overprom	utility
full_v5	0.720	0.067	0.263	0.122	0.337
no_calibration	0.679	0.087	0.265	0.164	0.244
no_burden_value	0.678	0.077	0.500	0.135	0.132
no_repair	0.671	0.074	0.258	0.133	0.259
no_overpromise	0.665	0.145	0.257	0.219	0.150
no_role	0.645	0.088	0.295	0.137	0.198
no_cap_map	0.633	0.139	0.255	0.226	0.118
v4_rules	0.625	0.109	0.293	0.180	0.126
no_intent	0.622	0.077	0.244	0.130	0.179
shared_only	0.601	0.130	0.264	0.210	0.070

Removing calibration, burden-aware query value, active repair, over-promise risk, role negotiation, capability maps, or intent belief weakens either success or utility. The ablations are not equal: no-burden query value preserves some success but pays large human burden, while no-overpromise risk degrades safety. That separation is the local mechanism evidence.

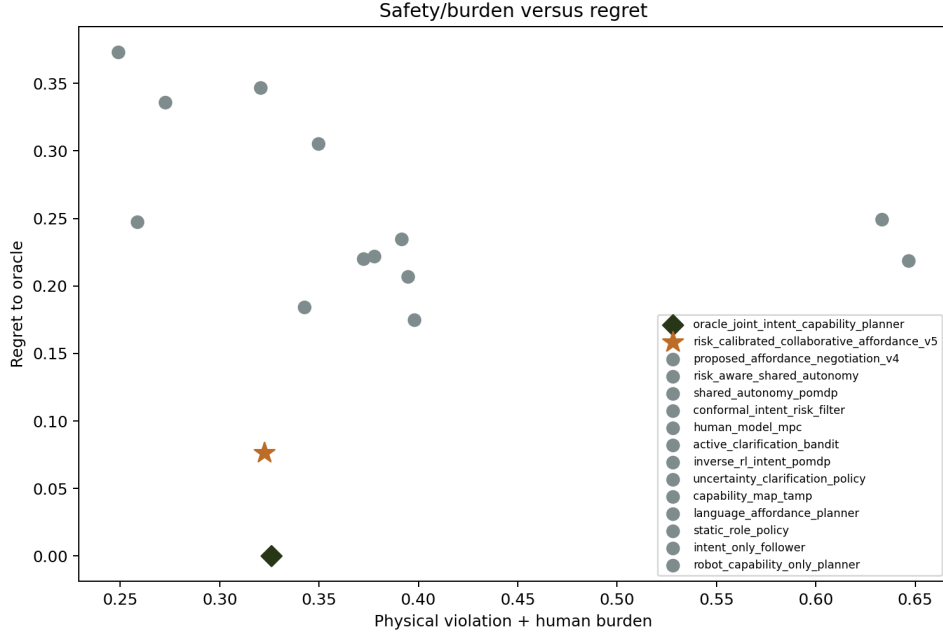


Figure 3: Physical violation plus human burden versus regret to oracle. V5 moves toward oracle without collapsing into clarification-only burden.

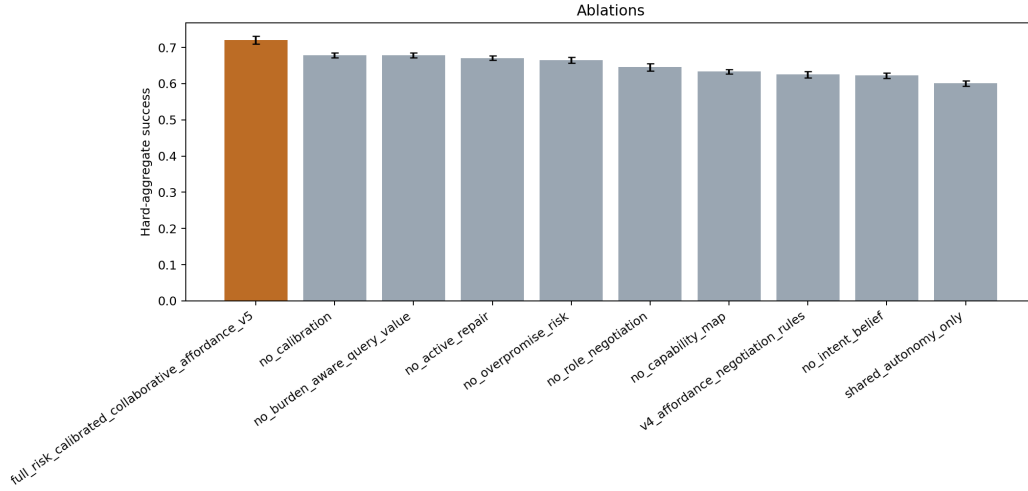


Figure 4: Ablation outcomes over hard collaborative-affordance splits.

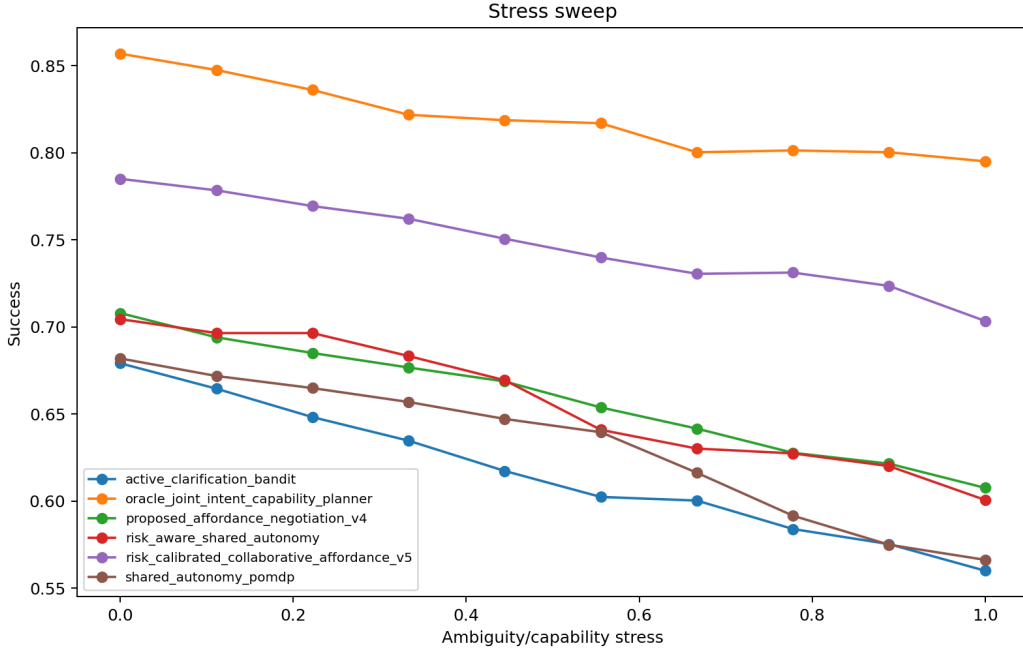


Figure 5: Maximum-stress success and utility pressure.

11 STRESS SWEEP

method	success	phys	burden	utility
oracle	0.795	0.096	0.342	0.317
collab_v5	0.703	0.089	0.323	0.222
risk_shared	0.601	0.150	0.283	0.012
collab_v4	0.608	0.154	0.344	-0.008
cap_tamp	0.515	0.142	0.185	-0.052
conformal_intent	0.551	0.145	0.331	-0.062
shared_pomdp	0.566	0.167	0.312	-0.079
human_mpc	0.555	0.201	0.270	-0.103
irl_intent	0.539	0.203	0.265	-0.128
active_clarify	0.560	0.173	0.580	-0.245

At maximum stress, the benchmark combines intent ambiguity, capability mismatch, delayed feedback, burden pressure, false clarification pressure, and overtrust safety pressure. V5 remains ahead of the strongest non-oracle method, but the oracle gap remains meaningful.

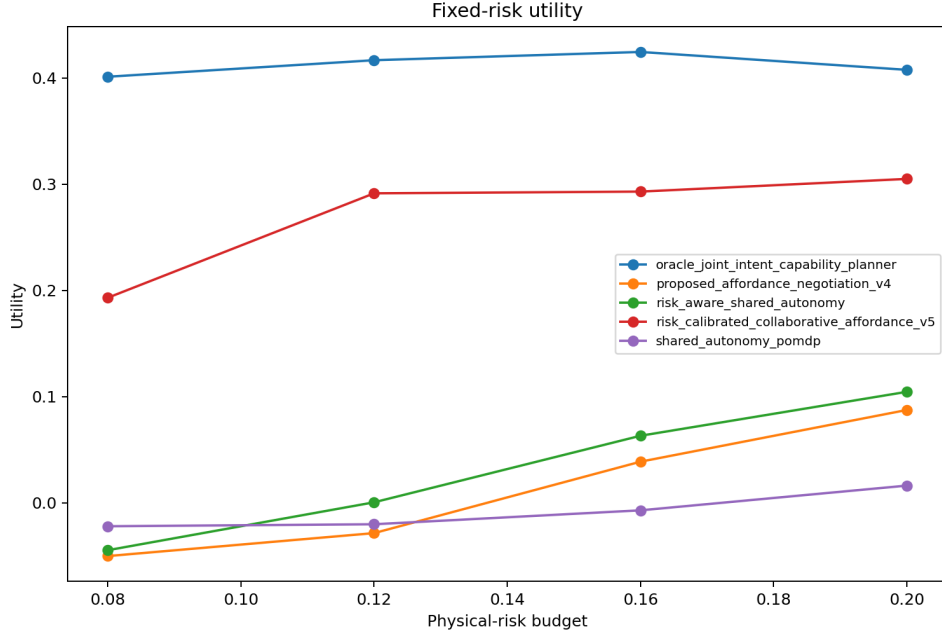


Figure 6: Fixed-risk utility. V5 does not win by abstaining: strict-budget coverage is retained by safe repair at a visible burden/query cost.

12 FIXED-RISK DEPLOYMENT

method	cover	success	phys	burden	unnecQ	utility
oracle	1.000	0.790	0.063	0.280	0.003	0.401
collab_v5	0.871	0.623	0.049	0.343	0.155	0.193
language_aff	0.006	0.003	0.000	0.006	0.006	-0.004
uncert_clarify	0.020	0.011	0.000	0.020	0.020	-0.009
cap_tamp	0.043	0.023	0.000	0.043	0.043	-0.020
shared_pomdp	0.056	0.034	0.000	0.056	0.056	-0.022
irl_intent	0.054	0.029	0.000	0.054	0.054	-0.026
conformal_intent	0.140	0.078	0.002	0.110	0.097	-0.036
human_mpc	0.086	0.047	0.000	0.086	0.086	-0.039
active_clarify	0.103	0.064	0.000	0.103	0.103	-0.040
risk_shared	0.119	0.070	0.000	0.117	0.117	-0.044
collab_v4	0.149	0.094	0.000	0.149	0.149	-0.050

The strict budget reveals the central tradeoff. V5 reaches fixed-risk coverage 0.871, but burden rises because safe repair/clarification is used instead of pretending high-risk actions are acceptable. This is the right shape of a local result: useful, but not yet a hardware safety claim.

13 FAILURE CASES

case	task	regime	split	gap	burden
1	co_carry	ergonomic_constraint	burden_shift	-0.667	0.333
2	assembly	occluded_goal	false_clarify	-0.500	0.500
3	sorting	ergonomic_constraint	combined_extreme	-0.500	0.833
4	handover	load_mismatch	combined_extreme	-0.667	0.667
5	tool_handoff	load_mismatch	combined_extreme	-0.667	0.167
6	handover	ambiguous_intent	burden_shift	-0.667	0.500
7	tool_handoff	ergonomic_constraint	overtrust	-0.500	0.500
8	assembly	commit_drift	combined_extreme	-0.500	0.500
9	co_carry	occluded_goal	false_clarify	-0.333	0.333
10	co_carry	safety_pref	false_clarify	-0.333	0.333
11	fixture_insert	safety_pref	burden_shift	-0.500	0.333
12	assembly	load_mismatch	false_clarify	-0.500	0.333

The negative cases concentrate where role negotiation and burden limits collide under delayed feedback. Keeping these cases in the paper prevents the result from becoming a pretty leaderboard with no failure model.

14 LIMITATIONS

This is a CPU-only local surrogate audit. It contains no real human-subject study, no real robot, no accepted high-fidelity collaborative-manipulation benchmark, no external benchmark, no calibrated external human-intent log, no trained checkpoint, and no rollout videos. The correct terminal state is **STRONG-REVISE**, not ICLR-main ready. The next real experiment must be externally frozen before tuning begins (Sanfilippo, 2024; Wang, 2023b; Ajoudani, 2023; Zou, 2022; Askarpour, 2020; K., 2022; Nikolaidis, 2024; Cheng, 2022; Marinelli, 2023; Lakshmaiya, 2025; Mehta, 2022; Hoffman, 2019; Wang, 2026a; Fave, 2023).

15 REPRODUCIBILITY

The main entry point is `src/run_experiment.py`. The manuscript is generated by `scripts/generate_manuscript.py`. The artifact validator checks row counts, finite metrics, page count, hash, artifact location, no Desktop copy, no repo-local numbered PDF, and boxed citation settings.

A FROZEN GATE LEDGER

All local empirical gates pass: success, physical safety, burden, intent, over-promise, conflict/query, calibration, regret, utility, pairwise, ablation, stress, and fixed-risk. The scope gate fails.

B FULL HARD-METRIC TABLE

method	succ	phys	burden	intent	overprom	ece	util
oracle	0.798	0.049	0.277	0.221	0.101	0.063	0.448
collab.v5	0.730	0.067	0.255	0.248	0.121	0.096	0.352
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uncert_clarify	0.557	0.143	0.490	0.304	0.231	0.275	-0.139
cap_tamp	0.553	0.101	0.157	0.401	0.166	0.211	0.095
language.aff	0.502	0.188	0.162	0.373	0.293	0.343	-0.106
static_role	0.466	0.175	0.098	0.424	0.266	0.331	-0.107
intent_only	0.460	0.230	0.091	0.373	0.341	0.394	-0.181
cap_only	0.440	0.160	0.089	0.481	0.236	0.315	-0.132

C CITATION LEDGER

id	theme	citations
1	physical human-robot collaboration	(Liu, 2019; Zhu, 2022; Steil, 2022)
2	shared autonomy and human intent inference	(Matsubara, 2022; Loianno, 2022b; Zhang, 2025c)
3	affordances, roles, and collaborative manipulation	(Wang, 2025; Thomaz, 2018; Peternel, 2025)
4	risk-aware planning and safety filters	(Loianno, 2022a; Zhu, 2021; Cangelosi, 2022)
5	active clarification and communication	(von Stryk; Spyros Maniatopoulos; Hadas KressGazit; Philipp Schillinger; David C. Conner, 2017; Ugur, 2019; Mihaylova, 2023)
6	human burden, trust, and ergonomics	(Rocco, 2022; OMalley, 2018; Straub, 2016)
7	evaluation, calibration, and reproducibility	(Germani, 2022; Zhang, 2017; Lanzotti, 2021)
8	physical human-robot collaboration	(N, 2023; Ceglarek, 2023; Phillips, 2021)
9	shared autonomy and human intent inference	(Monga, 2022; Ajoudani, 2022; Cheong, 2024)
10	affordances, roles, and collaborative manipulation	(Teixeira, 2025; Gibaru, 2022; Bilberg, 2019)
11	risk-aware planning and safety filters	(der Loos; Elizabeth A. Croft, 2020; van de Venn, 2022; Radwin, 2022)
12	active clarification and communication	(Rahman, 2017; Makris, 2022; Lee, 2022)
13	human burden, trust, and ergonomics	(Zhang, 2022; Francalanza, 2024; Jia, 2018)
14	evaluation, calibration, and reproducibility	(Lin, 2019; Pieters, 2021; Bonci, 2025)
15	physical human-robot collaboration	(Conforto, 2024; Sihn, 2017; Wang, 2022)
16	shared autonomy and human intent inference	(Peshkin, 2008; Hirche, 2020; Lohse, 2020)
17	affordances, roles, and collaborative manipulation	(Zeng, 2026; van der Spaa; Jens Kober; Michael Gienger, 2024; Braghin, 2022)
18	risk-aware planning and safety filters	(Makris, 2018; Elkmann, 2013; Wang, 2026b)
19	active clarification and communication	(Ajoudani, 2026; Scassellati, 2018; Salvietti, 2021)
20	human burden, trust, and ergonomics	(Arai, 2011; Sanfilippo, 2024; Wang, 2023b)
21	evaluation, calibration, and reproducibility	(Ajoudani, 2023; Zou, 2022; Askarpour, 2020)
22	physical human-robot collaboration	(K., 2022; Nikolaidis, 2024; Cheng, 2022)
23	shared autonomy and human intent inference	(Marinelli, 2023; Lakshmaia, 2025; Mehta, 2022)
24	affordances, roles, and collaborative manipulation	(Hoffman, 2019; Wang, 2026a; Fave, 2023)
25	risk-aware planning and safety filters	(Albayrak, 2021; Miranda, 2022; Arrichiello, 2021)
26	active clarification and communication	(de Moura Costa; Marcelo Roberto Petry; Antonio Paulo Moreira, 2022; Papakostas, 2022; Nassehi, 2022)
27	human burden, trust, and ergonomics	(Angelis, 2024; Franceschini, 2022; Lv, 2024)
28	evaluation, calibration, and reproducibility	(Boopathi, 2024; Garcia, 2020; Vidoni, 2021a)
29	physical human-robot collaboration	(Srinivasa, 2012; Roveda, 2024; Battini, 2024)
30	shared autonomy and human intent inference	(Fassi, 2025; Santos, 2021; Lohse, 2024)
31	affordances, roles, and collaborative manipulation	(Natale, 2021; Johansson, 2023; Gasparetto, 2022)
32	risk-aware planning and safety filters	(Makris, 2024; Padovano, 2022; Barua, 2025)
33	active clarification and communication	(Ryu, 2020; Yang, 2024; Srinivasa, 2018)
34	human burden, trust, and ergonomics	(Vidoni, 2021b; Pagilla, 2021; Leonessa, 2023)
35	evaluation, calibration, and reproducibility	(Tanie, 2003a; Wang, 2024; Rahman, 2019)
36	physical human-robot collaboration	(Aboeisheh, 2025; Pham, 2019b; Behdad, 2023)
37	shared autonomy and human intent inference	(van Dijk; Saskia J. Baltrusch; Ezra Dessers; Michiel P. de Looze, 2023; Zhang, 2023; Ravichandar, 2020)
38	affordances, roles, and collaborative manipulation	(Maged, 2025; Wittie, 2023; Gastaldi, 2020)
39	risk-aware planning and safety filters	(Germani, 2024; Sierra, 2017; Zhao, 2023)
40	active clarification and communication	(Ji, 2024; Srinivasa, 2017; Piga, 2022)
41	human burden, trust, and ergonomics	(Carpanzano, 2023; Angulo, 2021b; Suleiman, 2023)
42	evaluation, calibration, and reproducibility	(Kamat, 2019; Mendes, 2023; Makris, 2019)
43	physical human-robot collaboration	(Zhou; Steimle, 2021; Croft, 2025)
44	shared autonomy and human intent inference	(Pang, 2021; Lee, 2023; Pace, 2022)
45	affordances, roles, and collaborative manipulation	(de Koster; Seyyed Mahdi Ghorashi Khalilabadi, 2023; Petric, 2023; Marstio, 2018)
46	risk-aware planning and safety filters	(Martinez, 2022; Ansari, 2021; Wang, 2017a)
47	active clarification and communication	(ref, 2024; Ozceylan, 2020; Rausch, 2021)
48	human burden, trust, and ergonomics	(Otto, 2021; Krenn, 2023; Wang, 2019)
49	evaluation, calibration, and reproducibility	(Bilberg, 2018; Merdan, 2024; Jia, 2019)
50	physical human-robot collaboration	(Ma, 2025; Hace, 2023; Salmon, 2025)
51	shared autonomy and human intent inference	(Erickson, 2025; Chowdhury, 2023; Ruwinkel, 2021)
52	affordances, roles, and collaborative manipulation	(Reyes-Avendano, 2022; Zeiner, 2018; Wang, 2023a)
53	risk-aware planning and safety filters	(Franke, 2021; Liu, 2021; Tang, 2018)
54	active clarification and communication	(Angulo, 2021a; Pham, 2017; Fraisse, 2016)
55	human burden, trust, and ergonomics	(Basdogan, 2020; Bobovsky, 2021; Yang, 2021)
56	evaluation, calibration, and reproducibility	(de Souza; Elena Forsman; Enyu Cao; Shaolei Wang, 2018; Basdogan, 2018; Yang, 2022)
57	physical human-robot collaboration	(Wang, 2017b; Karimoddini, 2024; Jebelli, 2022)
58	shared autonomy and human intent inference	(Pintelon, 2024; Makris, 2021; Scassellati, 2022)
59	affordances, roles, and collaborative manipulation	(Zhou, 2019; Saif, 2020; Kamat, 2020)
60	risk-aware planning and safety filters	(Tanie, 2003b; Qiao, 2020; van de Venn; Hossein Karimpour, 2020)

61	active clarification and communication	(Lewis, 2020; Sabbatini, 2021; Christiansen, 2024)
62	human burden, trust, and ergonomics	(Negenborn, 2021; Balconi, 2025; Djerroud, 2025)
63	evaluation, calibration, and reproducibility	(Goerick, 2021; Piga, 2023b; Lewis, 2015)
64	physical human-robot collaboration	(Arrichiello, 2026; Zhang, 2025a; Kumaraguru, 2025)
65	shared autonomy and human intent inference	(Secchi, 2021; Lin, 2022; Gupta, 2013)
66	affordances, roles, and collaborative manipulation	(Torrás, 2016; Piga, 2023a; Pang, 2020)
67	risk-aware planning and safety filters	(Grieco, 2022; Fukuda, 2013; Elkmann, 2011)
68	active clarification and communication	(Mezouar, 2019; Stadnicka, 2019; Zeller, 2022)
69	human burden, trust, and ergonomics	(Kim, 2020; Radwin, 2024; Wang, 2018)
70	evaluation, calibration, and reproducibility	(Pham, 2019a; Yang, 2018; Biswas, 2024)
71	physical human-robot collaboration	(Marmaras, 2025; Sternad, 2017; Arai, 2009)
72	shared autonomy and human intent inference	(Secchi, 2019; Kopp, 2025; Scassellati, 2017)
73	affordances, roles, and collaborative manipulation	(Luca, 2013; Rad, 2024; Jiang, 2019)
74	risk-aware planning and safety filters	(Morgan, 2021; Zhang, 2025b; Wijesundara, 2018)
75	active clarification and communication	(Ahmad, 2026; Yldz, 2019; Fox, 2020)
76	human burden, trust, and ergonomics	(Urrea, 2025; Bottin, 2019; Bonk, 2023)
77	evaluation, calibration, and reproducibility	(Menezes, 2024; Ekstrom, 2020)

REFERENCES

- Affordance reasoning-based sequence planning manner for human-robot collaborative disassembly. *Journal of Mechanical Engineering*, 2024. doi: 10.3901/jme.2024.17.297. URL <https://doi.org/10.3901/jme.2024.17.297>.
- Raed Awashreh; Ahmed Aboeisheh. The collaborative future of translation between human-ai partnerships. *Advances in Computational Intelligence and Robotics Role of AI in Translation and Interpretation*, 2025. doi: 10.4018/979-8-3373-0060-3.ch008. URL <https://doi.org/10.4018/979-8-3373-0060-3.ch008>.
- Abdul Wali Khan; Maaz Akhtar; Sheheryar Mohsin Qureshi; Muzzamil Mustafa; Naser A. Al-saleh; Imran Ahmad. A systematic review of safety-driven approaches in humanrobot collaborative systems. *Sensors*, 2026. doi: 10.3390/s26072079. URL <https://doi.org/10.3390/s26072079>.
- Gustavo Jose Giardini Lahr; Doganay Sirintuna; Francesco Tassi; Heni Ben Amor; Arash Ajoudani. A non-parametric approach to exploring and quantifying the information flow in humanrobot collaboration. *ACM Transactions on Human-Robot Interaction*, 2026. doi: 10.1145/3744755. URL <https://doi.org/10.1145/3744755>.
- Marta Lagomarsino; Marta Lorenzini; Elena De Momi; Arash Ajoudani. Robot trajectory adaptation to optimise the trade-off between human cognitive ergonomics and workplace productivity in collaborative tasks. *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022. doi: 10.1109/iros47612.2022.9981424. URL <https://doi.org/10.1109/iros47612.2022.9981424>.
- Marta Lorenzini; Marta Lagomarsino; Luca Fortini; Soheil Gholami; Arash Ajoudani. Ergonomic human-robot collaboration in industry: A review. *Frontiers in Robotics and AI*, 2023. doi: 10.3389/frobt.2022.813907. URL <https://doi.org/10.3389/frobt.2022.813907>.
- O. Can Gorur; Benjamin Rosman; Fikret Sivrikaya; Sahin Albayrak. Fabric: A framework for the design and evaluation of collaborative robots with extended human adaptation. *arXiv*, 2021. doi: 10.1145/3585276. URL <http://arxiv.org/abs/2104.01976v2>.
- Luca Pietrantoni; Marco Favilla; Federico Fraboni; Elvis Mazzoni; Sofia Morandini; Martina Benvenuti; Marco De Angelis. Integrating collaborative robots in manufacturing, logistics, and agriculture: Expert perspectives on technical, safety, and human factors. *Frontiers in Robotics and AI*, 2024. doi: 10.3389/frobt.2024.1342130. URL <https://doi.org/10.3389/frobt.2024.1342130>.
- Alejandro Chacon; Pere Ponsa; Cecilio Angulo. Cognitive interaction analysis in humanrobot collaboration using an assembly task. *Electronics*, 2021a. doi: 10.3390/electronics10111317. URL <https://doi.org/10.3390/electronics10111317>.

- Alejandro Chacon; Pere Ponsa; Cecilio Angulo. Usability study through a human-robot collaborative workspace experience. *Designs*, 2021b. doi: 10.3390/designs5020035. URL <https://doi.org/10.3390/designs5020035>.
- Tadele Belay Tuli; Linus Kohl; Sisay Adugna Chala; Martin Manns; Fazel Ansari. Knowledge-based digital twin for predicting interactions in human-robot collaboration. 2021. doi: 10.1109/etfa45728.2021.9613342. URL <https://doi.org/10.1109/etfa45728.2021.9613342>.
- Jeffrey Too Chuan Tan; Feng Duan; Ye Zhang; Kei Watanabe; Ryu Kato; Tamio Arai. Human-robot collaboration in cellular manufacturing: Design and development. 2009. doi: 10.1109/iros.2009.5354155. URL <https://doi.org/10.1109/iros.2009.5354155>.
- Jeffrey Too Chuan Tan; Tamio Arai. Triple stereo vision system for safety monitoring of human-robot collaboration in cellular manufacturing. 2011. doi: 10.1109/isam.2011.5942335. URL <https://doi.org/10.1109/isam.2011.5942335>.
- Martina Lippi; Giuseppe Gillini; Alessandro Marino; Filippo Arrichiello. A data-driven approach for contact detection, classification and reaction in physical human-robot collaboration. *arXiv*, 2021. URL <http://arxiv.org/abs/2106.06781v1>.
- Mohammad Jahani Moghaddam; Filippo Arrichiello. Adaptive variable admittance control for intent-aware humanrobot collaboration. *Machines*, 2026. doi: 10.3390/machines14020221. URL <https://doi.org/10.3390/machines14020221>.
- Mehrnoosh Askarpour. How to formally model human in collaborative robotics. *Electronic Proceedings in Theoretical Computer Science*, 2020. doi: 10.4204/eptcs.329.1. URL <https://doi.org/10.4204/eptcs.329.1>.
- Davide Crivelli; Michela Balconi. Industrial collaborative robotics, humanoid cobots, and social embodiment: A critical look on humanrobot interaction. *International Journal of Humanoid Robotics*, 2025. doi: 10.1142/s0219843624300010. URL <https://doi.org/10.1142/s0219843624300010>.
- Zahir Babar; Rajesh Paul; Md Arifur Rahman; Tonmoy Barua. A systematic review of human-ai collaboration in it support services: Enhancing user experience and workflow automation. 2025. doi: 10.63125/grqtf978. URL <https://doi.org/10.63125/grqtf978>.
- Yusuf Aydn; Ozan Tokatl; Volkan Patoglu; Cagatay Basdogan. Stable physical human-robot interaction using fractional order admittance control. *IEEE Transactions on Haptics*, 2018. doi: 10.1109/toh.2018.2810871. URL <https://doi.org/10.1109/toh.2018.2810871>.
- Yusuf Aydn; Ozan Tokatl; Volkan Patoglu; Cagatay Basdogan. A computational multicriteria optimization approach to controller design for physical human-robot interaction. *IEEE Transactions on Robotics*, 2020. doi: 10.1109/tro.2020.2998606. URL <https://doi.org/10.1109/tro.2020.2998606>.
- Ali Keshvarparast; Nicola Berti; Saahil Chand; Mattia Guidolin; Yuqian Lu; Olga Battaia; Xun Xu; Daria Battini. Ergonomic design of human-robot collaborative workstation in the era of industry 5.0. *Computers & Industrial Engineering*, 2024. doi: 10.1016/j.cie.2024.110729. URL <https://doi.org/10.1016/j.cie.2024.110729>.
- Xinyao Zhang; Sibao Tian; Xiao Liang; Minghui Zheng; Sara Behdad. Early prediction of human intention for humanrobot collaboration using transformer network. *Journal of Computing and Information Science in Engineering*, 2023. doi: 10.1115/1.4064258. URL <https://doi.org/10.1115/1.4064258>.
- Ali Ahmad Malik; Arne Bilberg. Digital twins of human robot collaboration in a production setting. *Procedia Manufacturing*, 2018. doi: 10.1016/j.promfg.2018.10.047. URL <https://doi.org/10.1016/j.promfg.2018.10.047>.
- Ali Ahmad Malik; Arne Bilberg. Complexity-based task allocation in human-robot collaborative assembly. *Industrial Robot: the international journal of robotics research and application*, 2019. doi: 10.1108/ir-11-2018-0231. URL <https://doi.org/10.1108/ir-11-2018-0231>.

- Mukund Mitra; Gyanig Kumar; Partha Pratim Chakrabarti; Pradipta Biswas. Enhanced human-robot collaboration with intent prediction using deep inverse reinforcement learning. *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024. doi: 10.1109/icra57147.2024.10610595. URL <https://doi.org/10.1109/icra57147.2024.10610595>.
- Stefan Grushko; Ales Vysocky; Petr Oscadal; Michal Vocetka; Petr Novak; Zdenko Bobovsky. Improved mutual understanding for human-robot collaboration: Combining human-aware motion planning with haptic feedback devices for communicating planned trajectory. *Sensors*, 2021. doi: 10.3390/s21113673. URL <https://doi.org/10.3390/s21113673>.
- Renat Kermenov; Alessandro Di Biase; Ilaria Pellicani; Sauro Longhi; Andrea Bonci. Near time-optimal trajectories with iso standard constraints for humanrobot collaboration in fabric co-transportation. *Robotics*, 2025. doi: 10.3390/robotics14020010. URL <https://doi.org/10.3390/robotics14020010>.
- Ronghuai Huang; Ahmed Tlili; Lin Xu; Chen Ying; Lanqin Zheng; Ahmed Hosny Saleh Metwally; Da Ting; TingWen Chang; Huanhuan Wang; Jon Mason; Christian M. Stracke; Demetrios G. Sampson; Curtis J. Bonk. Educational futures of intelligent synergies between humans, digital twins, avatars, and robots - the istar framework. *Journal of Applied Learning & Teaching*, 2023. doi: 10.37074/jalt.2023.6.2.33. URL <https://doi.org/10.37074/jalt.2023.6.2.33>.
- Tejaswini A. Puranik; Nazeer Shaik; Ramu Vankudoth; Mohini R. Kolhe; Neelam Yadav; Sampath Boopathi. Study on harmonizing human-robot (drone) collaboration. *Advances in information security, privacy, and ethics book series*, 2024. doi: 10.4018/979-8-3693-0774-8.ch001. URL <https://doi.org/10.4018/979-8-3693-0774-8.ch001>.
- Maurizio Faccio; Riccardo Minto; Giulio Rosati; Matteo Bottin. The influence of the product characteristics on human-robot collaboration: a model for the performance of collaborative robotic assembly. *The International Journal of Advanced Manufacturing Technology*, 2019. doi: 10.1007/s00170-019-04670-6. URL <https://doi.org/10.1007/s00170-019-04670-6>.
- Marco Maccarini; Filippo Pura; Dario Piga; Loris Roveda; LG Mantovani; Francesco Braghin. Preference-based optimization of a human-robot collaborative controller. *IFAC-PapersOnLine*, 2022. doi: 10.1016/j.ifacol.2023.01.126. URL <https://doi.org/10.1016/j.ifacol.2023.01.126>.
- Francesco Semeraro; Alexander Griffiths; Angelo Cangelosi. Humanrobot collaboration and machine learning: A systematic review of recent research. *Robotics and Computer-Integrated Manufacturing*, 2022. doi: 10.1016/j.rcim.2022.102432. URL <https://doi.org/10.1016/j.rcim.2022.102432>.
- Elias Montini; Vincenzo Cutrona; Samuele DellOca; Giuseppe Landolfi; Andrea Bettoni; Paolo Rocco; Emanuele Carpanzano. A framework for human-aware collaborative robotics systems development. *Procedia CIRP*, 2023. doi: 10.1016/j.procir.2023.09.129. URL <https://doi.org/10.1016/j.procir.2023.09.129>.
- Yuzhu Sun; Mien Van; Stephen McIlvanna; Nhat Nguyen Minh; Sean McLoone; Dariusz Ceglarek. Adaptive admittance control for safety-critical physical human robot collaboration. *IFAC-PapersOnLine*, 2023. doi: 10.1016/j.ifacol.2023.10.1772. URL <https://doi.org/10.1016/j.ifacol.2023.10.1772>.
- Simon Armleder; Emmanuel Dean-Leon; Florian Bergner; Gordon Cheng. Interactive force control based on multimodal robot skin for physical humanrobot collaboration. *Advanced Intelligent Systems*, 2022. doi: 10.1002/aisy.202100047. URL <https://doi.org/10.1002/aisy.202100047>.
- Dawoon Jung; Chengyan Gu; Junmin Park; Joono Cheong. Touch gesture recognition-based physical humanrobot interaction for collaborative tasks. *IEEE Transactions on Cognitive and Developmental Systems*, 2024. doi: 10.1109/tcds.2024.3466553. URL <https://doi.org/10.1109/tcds.2024.3466553>.

- Hribhu Chowdhury. Human-robot collaboration in manufacturing assembly tasks. *Preprints.org*, 2023. doi: 10.20944/preprints202310.0049.v2. URL <https://doi.org/10.20944/preprints202310.0049.v2>.
- Ilona Buchem; Stefano Sostak; Lewe Christiansen. Humanrobot co-facilitation in collaborative learning: A comparative study of the effects of human and robot facilitation on learning experience and learning outcomes. *J*, 2024. doi: 10.3390/j7030014. URL <https://doi.org/10.3390/j7030014>.
- Alessia Nobile; Daniele Bibbo; Marta Russo; Silvia Conforto. A focus on quantitative methods to assess human factors in collaborative robotics. *International Journal of Industrial Ergonomics*, 2024. doi: 10.1016/j.ergon.2024.103663. URL <https://doi.org/10.1016/j.ergon.2024.103663>.
- Wesley P. Chan; Morgan Crouch; Khoa Hoang; Charlie Chen; Nicole Robinson; Elizabeth Croft. Improving humanrobot collaboration through augmented reality and eye gaze. *ACM Transactions on Human-Robot Interaction*, 2025. doi: 10.1145/3716175. URL <https://doi.org/10.1145/3716175>.
- Kaveh Azadeh; Debjit Roy; Rene de Koster; Seyyed Mahdi Ghorashi Khalilabadi. Zoning strategies for humanrobot collaborative picking. *Decision Sciences*, 2023. doi: 10.1111/deci.12620. URL <https://doi.org/10.1111/deci.12620>.
- Gabriel de Moura Costa; Marcelo Roberto Petry; Antonio Paulo Moreira. Augmented reality for humanrobot collaboration and cooperation in industrial applications: A systematic literature review. *Sensors*, 2022. doi: 10.3390/s22072725. URL <https://doi.org/10.3390/s22072725>.
- Rafia Inam; Klaus Raizer; Alberto Hata; Ricardo Gabbay de Souza; Elena Forsman; Enyu Cao; Shaolei Wang. Risk assessment for human-robot collaboration in an automated warehouse scenario. 2018. doi: 10.1109/etfa.2018.8502466. URL <https://doi.org/10.1109/etfa.2018.8502466>.
- Wesley P. Chan; Geoffrey Hanks; Maram Sakr; Tiger Zuo; H. F. Machiel Van der Loos; Elizabeth A. Croft. An augmented reality human-robot physical collaboration interface design for shared, large-scale, labour-intensive manufacturing tasks. 2020. doi: 10.1109/iro45743.2020.9341119. URL <https://doi.org/10.1109/iro45743.2020.9341119>.
- Mokone Kekana; Shengzhi Du; Nico Steyn; Abderraouf Benali; Halim Djerroud. A review of human intention recognition frameworks in industrial collaborative robotics. *Robotics*, 2025. doi: 10.3390/robotics14120174. URL <https://doi.org/10.3390/robotics14120174>.
- Atieh Hanna; Kristofer Bengtsson; Per-Lage Gotvall; Mikael Ekstrom. Towards safe human robot collaboration - risk assessment of intelligent automation. 2020. doi: 10.1109/etfa46521.2020.9212127. URL <https://doi.org/10.1109/etfa46521.2020.9212127>.
- Christian Vogel; Christoph Walter; Norbert Elkmann. A projection-based sensor system for safe physical human-robot collaboration. 2013. doi: 10.1109/iro45743.2020.9341119. URL <https://doi.org/10.1109/iro45743.2020.9341119>.
- Christian Vogel; Maik Poggendorf; Christoph Walter; Norbert Elkmann. Towards safe physical human-robot collaboration: A projection-based safety system. *2011 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2011. doi: 10.1109/iro45743.2020.9341119. URL <https://doi.org/10.1109/iro45743.2020.9341119>.
- Junxiang Wang; Emek Bars Kucuktabak; Rana Soltani Zarrin; Zackory Erickson. Cori: Communication of robot intent for physical human-robot interaction. *arXiv*, 2025. URL <http://arxiv.org/abs/2505.20537v2>.
- Samarathunga S. M. B. P. B.; Marcello Valori; Giovanni Legnani; Irene Fassi. Assessing safety in physical humanrobot interaction in industrial settings: A systematic review of contact modelling and impact measuring methods. *Robotics*, 2025. doi: 10.3390/robotics14030027. URL <https://doi.org/10.3390/robotics14030027>.

- Claudia Carissoli; Luca Negri; Marta Bassi; Fabio Alexander Storm; Antonella Delle Fave. Mental workload and human-robot interaction in collaborative tasks: A scoping review. *International Journal of Human-Computer Interaction*, 2023. doi: 10.1080/10447318.2023.2254639. URL <https://doi.org/10.1080/10447318.2023.2254639>.
- Wei Yang; Chris Paxton; Maya Cakmak; Dieter Fox. Human grasp classification for reactive human-to-robot handovers. 2020. doi: 10.1109/iros45743.2020.9341004. URL <https://doi.org/10.1109/iros45743.2020.9341004>.
- Andrea Cherubini; Robin Passama; Andre Crosnier; Antoine Lasnier; Philippe Fraisse. Collaborative manufacturing with physical humanrobot interaction. *Robotics and Computer-Integrated Manufacturing*, 2016. doi: 10.1016/j.rcim.2015.12.007. URL <https://doi.org/10.1016/j.rcim.2015.12.007>.
- Amberlynn Bonello; Paul Refalo; Emmanuel Francalanza. The impacts of industrial safety on environmental sustainability in human-robot-collaboration within industry 5.0. *Procedia CIRP*, 2024. doi: 10.1016/j.procir.2024.01.041. URL <https://doi.org/10.1016/j.procir.2024.01.041>.
- Riccardo Gervasi; Khurshid Aliev; Luca Mastrogiacomo; Fiorenzo Franceschini. User experience and physiological response in human-robot collaboration: A preliminary investigation. *Journal of Intelligent & Robotic Systems*, 2022. doi: 10.1007/s10846-022-01744-8. URL <https://doi.org/10.1007/s10846-022-01744-8>.
- Paul Heisler; Daniel Utsch; Marlene Kuhn; Jorg Franke. Optimization of wire harness assembly using humanrobot-collaboration. *Procedia CIRP*, 2021. doi: 10.1016/j.procir.2020.05.235. URL <https://doi.org/10.1016/j.procir.2020.05.235>.
- Fei Chen; Kosuke Sekiyama; Ferdinando Cannella; Toshio Fukuda. Optimal subtask allocation for human and robot collaboration within hybrid assembly system. *IEEE Transactions on Automation Science and Engineering*, 2013. doi: 10.1109/tase.2013.2274099. URL <https://doi.org/10.1109/tase.2013.2274099>.
- Carlos A. Garcia; William Montalvo; Marcelo V. Garcia. Human-robot collaboration based on cyber-physical production system and mqtt. *Procedia Manufacturing*, 2020. doi: 10.1016/j.promfg.2020.02.088. URL <https://doi.org/10.1016/j.promfg.2020.02.088>.
- Lorenzo Scalera; Andrea Giusti; Renato Vidoni; Alessandro Gasparetto. Enhancing fluency and productivity in human-robot collaboration through online scaling of dynamic safety zones. *The International Journal of Advanced Manufacturing Technology*, 2022. doi: 10.1007/s00170-022-09781-1. URL <https://doi.org/10.1007/s00170-022-09781-1>.
- Elisa Digo; Mattia Antonelli; Valerio Cornagliotto; Stefano Pastorelli; Laura Gastaldi. Collection and analysis of human upper limbs motion features for collaborative robotic applications. *Robotics*, 2020. doi: 10.3390/robotics9020033. URL <https://doi.org/10.3390/robotics9020033>.
- Marianna Ciccarelli; Alessandra Papetti; Cecilia Scoccia; Giacomo Menchi; Leonardo Mostarda; Giacomo Palmieri; Michele Germani. A system to improve the physical ergonomics in human-robot collaboration. *Procedia Computer Science*, 2022. doi: 10.1016/j.procs.2022.01.267. URL <https://doi.org/10.1016/j.procs.2022.01.267>.
- Marianna Ciccarelli; Matteo Forlini; Alessandra Papetti; Giacomo Palmieri; Michele Germani. Advancing humanrobot collaboration in handcrafted manufacturing: cobot-assisted polishing design boosted by virtual reality and human-in-the-loop. *The International Journal of Advanced Manufacturing Technology*, 2024. doi: 10.1007/s00170-024-13639-z. URL <https://doi.org/10.1007/s00170-024-13639-z>.
- Anthony Quenehen; Nathalie Klement; Amine Mohamed Abdeljaouad; Lionel Roucoules; Olivier Gibaru. Economic and ergonomic performance enhancement in assembly process through multiple collaboration modes between human and robot. *International Journal of Production Research*, 2022. doi: 10.1080/00207543.2022.2039795. URL <https://doi.org/10.1080/00207543.2022.2039795>.

- Samuele Vinanzi; Angelo Cangelosi; Christian Goerick. The collaborative mind: intention reading and trust in human-robot interaction. *iScience*, 2021. doi: 10.1016/j.isci.2021.102130. URL <https://doi.org/10.1016/j.isci.2021.102130>.
- Jonathan Cacace; Riccardo Caccavale; Alberto Finzi; Riccardo Grieco. Combining human guidance and structured task execution during physical humanrobot collaboration. *Journal of Intelligent Manufacturing*, 2022. doi: 10.1007/s10845-022-01989-y. URL <https://doi.org/10.1007/s10845-022-01989-y>.
- Carlos Morato; Krishnanand N. Kaipa; Boxuan Zhao; Satyandra K. Gupta. Toward safe human robot collaboration by using multiple kinects based real-time human tracking. *Journal of Computing and Information Science in Engineering*, 2013. doi: 10.1115/1.4025810. URL <https://doi.org/10.1115/1.4025810>.
- Milan Zorman; Bojan Zlahtic; Sasa Stradovnik; Ales Hace. Transferring artificial intelligence practices between collaborative robotics and autonomous driving. *Kybernetes*, 2023. doi: 10.1108/k-05-2022-0679. URL <https://doi.org/10.1108/k-05-2022-0679>.
- Selma Music; Sandra Hirche. Haptic shared control for human-robot collaboration: A game-theoretical approach. *IFAC-PapersOnLine*, 2020. doi: 10.1016/j.ifacol.2020.12.2751. URL <https://doi.org/10.1016/j.ifacol.2020.12.2751>.
- Guy Hoffman. Evaluating fluency in humanrobot collaboration. *IEEE Transactions on Human-Machine Systems*, 2019. doi: 10.1109/thms.2019.2904558. URL <https://doi.org/10.1109/thms.2019.2904558>.
- Yizhi Liu; Houtan Jebelli. Intention estimation in physical human-robot interaction in construction: Empowering robots to gauge workers posture. *Construction Research Congress* 2022, 2022. doi: 10.1061/9780784483961.065. URL <https://doi.org/10.1061/9780784483961.065>.
- Yingchao You; Boliang Cai; Duc Truong Pham; Ying Liu; Ze Ji. A human digital twin approach for fatigue-aware task planning in human-robot collaborative assembly. *Computers & Industrial Engineering*, 2024. doi: 10.1016/j.cie.2024.110774. URL <https://doi.org/10.1016/j.cie.2024.110774>.
- Weitian Wang; Rui Li; Yi Chen; Z. Max Diekel; Yunyi Jia. Facilitating humanrobot collaborative tasks by teaching-learning-collaboration from human demonstrations. *IEEE Transactions on Automation Science and Engineering*, 2019. doi: 10.1109/tase.2018.2840345. URL <https://doi.org/10.1109/tase.2018.2840345>.
- Weitian Wang; Rui Li; Z. Max Diekel; Yi Chen; Zhujun Zhang; Yunyi Jia. Controlling object hand-over in humanrobot collaboration via natural wearable sensing. *IEEE Transactions on Human-Machine Systems*, 2018. doi: 10.1109/thms.2018.2883176. URL <https://doi.org/10.1109/thms.2018.2883176>.
- Khurram Monir Rabby; Mubbashar Altaf Khan; Ali Karimoddini; Steven Jiang. An effective model for human cognitive performance within a human-robot collaboration framework. 2019. doi: 10.1109/smc.2019.8914536. URL <https://doi.org/10.1109/smc.2019.8914536>.
- Omkar Salunkhe; Johan Stahre; David Romero; Dan Li; Bjorn Johansson. Specifying task allocation in automotive wire harness assembly stations for human-robot collaboration. *Computers & Industrial Engineering*, 2023. doi: 10.1016/j.cie.2023.109572. URL <https://doi.org/10.1016/j.cie.2023.109572>.
- Anil Kumar Inkulu; M.V.A. Raju Bahubalendruni; Ashok Dara; SankaranarayanaSamy K. Challenges and opportunities in human robot collaboration context of industry 4.0 - a state of the art review. *Industrial Robot: the international journal of robotics research and application*, 2022. doi: 10.1108/ir-04-2021-0077. URL <https://doi.org/10.1108/ir-04-2021-0077>.
- CiJyun Liang; Wes McGee; Carol C. Menassa; Vineet R. Kamat. Bi-directional communication bridge for state synchronization between digital twin simulations and physical construction robots. *Proceedings of the ... ISARC*, 2020. doi: 10.22260/isarc2020/0205. URL <https://doi.org/10.22260/isarc2020/0205>.

- Daeho Kim; Ankit Goyal; Alejandro Newell; Sang Hyun Lee; Jia Deng; Vineet R. Kamat. Semantic relation detection between construction entities to support safe human-robot collaboration in construction. 2019. doi: 10.1061/9780784482438.034. URL <https://doi.org/10.1061/9780784482438.034>.
- Md Khurram Monir Rabby; Mubbashar Altaf Khan; Steven Xiaochun Jiang; Ali Karimoddini. Performance-aware trust modeling within a humanmulti-robot collaboration setting. *ACM Transactions on Human-Robot Interaction*, 2024. doi: 10.1145/3660648. URL <https://doi.org/10.1145/3660648>.
- Hanjin Jo; Woong Choi; Geonhui Lee; Wookhyun Park; Jaehyo Kim. Analysis of visuo motor control between dominant hand and non-dominant hand for effective human-robot collaboration. *Sensors*, 2020. doi: 10.3390/s20216368. URL <https://doi.org/10.3390/s20216368>.
- Florian Schroder; Fabian Heinrich; Stefan Kopp. Towards fluid human-agent collaboration: From dynamic collaboration patterns to models of theory of mind reasoning. *Frontiers in Robotics and AI*, 2025. doi: 10.3389/frobt.2025.1532693. URL <https://doi.org/10.3389/frobt.2025.1532693>.
- Stephanie Gross; Brigitte Krenn. A communicative perspective on humanrobot collaboration in industry: Mapping communicative modes on collaborative scenarios. *International Journal of Social Robotics*, 2023. doi: 10.1007/s12369-023-00991-5. URL <https://doi.org/10.1007/s12369-023-00991-5>.
- Madhanagopal Manoharan; Ranga Rakesh Ranganayakulu; Senthilkumaran Kumaraguru. Affordable collaborative robot technology for human-robot collaboration applications. *Volume 4: Gas Turbines; Pressure Technologies; Robotics & Automation*, 2025. doi: 10.1115/imece-india2025-161095. URL <https://doi.org/10.1115/imece-india2025-161095>.
- Ronit Shah; Arockia Selvakumar Arockia Doss; Natrayan Lakshmaiya. Advancements in ai-enhanced collaborative robotics: towards safer, smarter, and human-centric industrial automation. *Results in Engineering*, 2025. doi: 10.1016/j.rineng.2025.105704. URL <https://doi.org/10.1016/j.rineng.2025.105704>.
- Andrea Rega; Castrese Di Marino; Agnese Pasquariello; Ferdinando Vitolo; Stanislao Patalano; Alessandro Zanella; Antonio Lanzotti. Collaborative workplace design: A knowledge-based approach to promote humanrobot collaboration and multi-objective layout optimization. *Applied Sciences*, 2021. doi: 10.3390/app112412147. URL <https://doi.org/10.3390/app112412147>.
- Hongrui Yu; Vineet R. Kamat; Carol C. Menassa; Wes McGee; Yijie Guo; Honglak Lee. Mutual physical state-aware object handover in full-contact collaborative human-robot construction work. *Automation in Construction*, 2023. doi: 10.1016/j.autcon.2023.104829. URL <https://doi.org/10.1016/j.autcon.2023.104829>.
- Matteo Pantano; Qiaoyue Yang; Adrian Blumberg; Raven Reisch; Tobias Hauser; Benjamin Lutz; Daniel Regulin; Tobias Kamps; Konstantinos Traganos; Dongheui Lee. Influence of task decision autonomy on physical ergonomics and robot performances in an industrial humanrobot collaboration scenario. *Frontiers in Robotics and AI*, 2022. doi: 10.3389/frobt.2022.943261. URL <https://doi.org/10.3389/frobt.2022.943261>.
- Satyajit Upasani; Divya Srinivasan; Qi Zhu; Jing Du; Alexander Leonessa. Eye-tracking in physical humanrobot interaction: Mental workload and performance prediction. *Human Factors The Journal of the Human Factors and Ergonomics Society*, 2023. doi: 10.1177/00187208231204704. URL <https://doi.org/10.1177/00187208231204704>.
- Isura Ranatunga; Sven Cremer; Dan O. Popa; Frank L. Lewis. Intent aware adaptive admittance control for physical human-robot interaction. 2015. doi: 10.1109/icra.2015.7139988. URL <https://doi.org/10.1109/icra.2015.7139988>.

- Sven Cremer; Sumit Kumar Das; Indika B. Wijayasinghe; Dan O. Popa; Frank L. Lewis. Model-free online neuroadaptive controller with intent estimation for physical humanrobot interaction. *IEEE Transactions on Robotics*, 2020. doi: 10.1109/tro.2019.2946721. URL <https://doi.org/10.1109/tro.2019.2946721>.
- Qiao Wang; Dikai Liu; Marc G. Carmichael; Stefano Aldini; Chin-Teng Lin. Computational model of robot trust in human co-worker for physical human-robot collaboration. *IEEE Robotics and Automation Letters*, 2022. doi: 10.1109/lra.2022.3145957. URL <https://doi.org/10.1109/lra.2022.3145957>.
- Stefano Aldini; Ashlesha Akella; Avinash Kumar Singh; YuKai Wang; Marc G. Carmichael; Dikai Liu; ChinTeng Lin. Effect of mechanical resistance on cognitive conflict in physical human-robot collaboration. 2019. doi: 10.1109/icra.2019.8793748. URL <https://doi.org/10.1109/icra.2019.8793748>.
- Ming Zhang; Chunquan Li; Yuling Shang; Hongyan Huang; Wangchun Zhu; Yujia Liu. A task scheduling model integrating micro-breaks for optimisation of job-cycle time in human-robot collaborative assembly cells. *International Journal of Production Research*, 2021. doi: 10.1080/00207543.2021.1937746. URL <https://doi.org/10.1080/00207543.2021.1937746>.
- Uchenna Emeoha Ogenyi; Jinguo Liu; Chenguang Yang; Zhaojie Ju; Honghai Liu. Physical humanrobot collaboration: Robotic systems, learning methods, collaborative strategies, sensors, and actuators. *IEEE Transactions on Cybernetics*, 2019. doi: 10.1109/tcyb.2019.2947532. URL <https://doi.org/10.1109/tcyb.2019.2947532>.
- Ali Al-Yacoub; Achim Buerkle; Myles Flanagan; Pedro Ferreira; EllaMae Hubbard; Niels Lohse. Effective human-robot collaboration through wearable sensors. 2020. doi: 10.1109/etfa46521.2020.9212100. URL <https://doi.org/10.1109/etfa46521.2020.9212100>.
- Tiziana C. Callari; Riccardo Vecellio Segate; Ella-Mae Hubbard; Angela Daly; Niels Lohse. An ethical framework for human-robot collaboration for the future people-centric manufacturing: A collaborative endeavour with european subject-matter experts in ethics. *Technology in Society*, 2024. doi: 10.1016/j.techsoc.2024.102680. URL <https://doi.org/10.1016/j.techsoc.2024.102680>.
- Guanrui Li; Xinyang Liu; Giuseppe Loianno. Human-aware physical human-robot collaborative transportation and manipulation with multiple aerial robots. *arXiv*, 2022a. doi: 10.1109/tro.2024.3502508. URL <http://arxiv.org/abs/2210.05894v3>.
- Guanrui Li; Xinyang Liu; Giuseppe Loianno. Human-aware physical human-robot collaborative transportation and manipulation with multiple aerial robots. *arXiv (Cornell University)*, 2022b. doi: 10.48550/arxiv.2210.05894. URL <https://doi.org/10.48550/arxiv.2210.05894>.
- Fabrizio Flacco; Alessandro De Luca. Safe physical human-robot collaboration. 2013. doi: 10.1109/iro.2013.6696635. URL <https://doi.org/10.1109/iro.2013.6696635>.
- Shanhe Lou; Yiran Zhang; Runjia Tan; Chen Lv. A human-cyber-physical system enabled sequential disassembly planning approach for a human-robot collaboration cell in industry 5.0. *Robotics and Computer-Integrated Manufacturing*, 2024. doi: 10.1016/j.rcim.2023.102706. URL <https://doi.org/10.1016/j.rcim.2023.102706>.
- Yang Liu; Fuqiang Guo; Yan Ma. A stackelberg trust-based humanrobot collaboration framework for warehouse picking. *Systems*, 2025. doi: 10.3390/systems13050348. URL <https://doi.org/10.3390/systems13050348>.
- Mina Yousry Halim; Mohammed Ibrahim Awad; Shady A. Maged. Hybrid physics-infused deep learning for enhanced real-time prediction of human upper limb movements in collaborative robotics. *Journal of Intelligent & Robotic Systems*, 2025. doi: 10.1007/s10846-025-02237-0. URL <https://doi.org/10.1007/s10846-025-02237-0>.

- Alessandro Umbrico; Andrea Orlandini; Amedeo Cesta; Marco Faroni; Manuel Beschi; Nicola Pedrocchi; Andrea Scala; Piervincenzo Tavormina; Spyridon Koukas; Andreas Zalonis; Nikos Fourtakas; Panagiotis Stylianos Kotsaris; Dionisis Andronas; Sotiris Makris. Design of advanced humanrobot collaborative cells for personalized humanrobot collaborations. *Applied Sciences*, 2022. doi: 10.3390/app12146839. URL <https://doi.org/10.3390/app12146839>.
- Konstantinos Katsampiris-Salgado; Nikos Dimitropoulos; Christos Gkrizis; George Michalos; Sotiris Makris. Advancing human-robot collaboration: Predicting operator trajectories through ai and infrared imaging. *Journal of Manufacturing Systems*, 2024. doi: 10.1016/j.jmsy.2024.05.015. URL <https://doi.org/10.1016/j.jmsy.2024.05.015>.
- Nikolaos Nikolakis; ; Sotiris Makris. A cyber-physical context-aware system for coordinating human-robot collaboration. *Procedia CIRP*, 2018. doi: 10.1016/j.procir.2018.03.033. URL <https://doi.org/10.1016/j.procir.2018.03.033>.
- Nikolaos Nikolakis; Vasilis Maratos; Sotiris Makris. A cyber physical system (cps) approach for safe human-robot collaboration in a shared workplace. *Robotics and Computer-Integrated Manufacturing*, 2019. doi: 10.1016/j.rcim.2018.10.003. URL <https://doi.org/10.1016/j.rcim.2018.10.003>.
- Nikos Dimitropoulos; ; George Michalos; Sotiris Makris. Operator support in humanrobot collaborative environments using ai enhanced wearable devices. *Procedia CIRP*, 2021. doi: 10.1016/j.procir.2020.07.006. URL <https://doi.org/10.1016/j.procir.2020.07.006>.
- Marina Marinelli. From industry 4.0 to construction 5.0: Exploring the path towards humanrobot collaboration in construction. *Systems*, 2023. doi: 10.3390/systems11030152. URL <https://doi.org/10.3390/systems11030152>.
- Loizos Psarakis; Dimitris Nathanael; Nicolas Marmaras. Communicating robots intent through visual cues enhances human anticipatory behavior in humandual robot collaboration. *Robotics and Computer-Integrated Manufacturing*, 2025. doi: 10.1016/j.rcim.2024.102886. URL <https://doi.org/10.1016/j.rcim.2024.102886>.
- Iina Aaltonen; Timo Salmi; Ilari Marstio. Refining levels of collaboration to support the design and evaluation of human-robot interaction in the manufacturing industry. *Procedia CIRP*, 2018. doi: 10.1016/j.procir.2018.03.214. URL <https://doi.org/10.1016/j.procir.2018.03.214>.
- Lina Maria Amaya-Mejia; Nicolas Duque-Suarez; Daniel Jaramillo-Ramirez; Carol Martinez. Vision-based safety system for barrierless human-robot collaboration. *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022. doi: 10.1109/iros47612.2022.9981689. URL <https://doi.org/10.1109/iros47612.2022.9981689>.
- Yujun Lai; Gavin Paul; Yunduan Cui; Takamitsu Matsubara. User intent estimation during robot learning using physical human robot interaction primitives. *Autonomous Robots*, 2022. doi: 10.1007/s10514-021-10030-9. URL <https://doi.org/10.1007/s10514-021-10030-9>.
- Sarah Hopko; Jingkun Wang; Ranjana Mehta. Human factors considerations and metrics in shared space human-robot collaboration: A systematic review. *Frontiers in Robotics and AI*, 2022. doi: 10.3389/frobt.2022.799522. URL <https://doi.org/10.3389/frobt.2022.799522>.
- Pedro P. Garcia; Telmo G. Santos; Miguel A. Machado; Nuno Mendes. Deep learning framework for controlling work sequence in collaborative humanrobot assembly processes. *Sensors*, 2023. doi: 10.3390/s23010553. URL <https://doi.org/10.3390/s23010553>.
- Jovan Clive Menezes. Mumodar: Multi-modal framework for human-robot collaboration in cyber-physical systems. *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*, 2024. doi: 10.1145/3610978.3640688. URL <https://doi.org/10.1145/3610978.3640688>.

- Timon Hoebert; Stephan Seibel; Manuel Amersdorfer; Markus Vincze; Wilfried Lopuschitz; Munir Merdan. A framework for enhanced humanrobot collaboration during disassembly using digital twin and virtual reality. *Robotics*, 2024. doi: 10.3390/robotics13070104. URL <https://doi.org/10.3390/robotics13070104>.
- Harsh Maithani; Juan Antonio Corrales Ramon; Youcef Mezouar. Predicting human intent for cooperative physical human-robot interaction tasks. 2019. doi: 10.1109/icca.2019.8899490. URL <https://doi.org/10.1109/icca.2019.8899490>.
- Shenglin Wang; Jingqiong Zhang; Peng Wang; James Law; Radu Calinescu; Lyudmila Mihaylova. A deep learning-enhanced digital twin framework for improving safety and reliability in human-robot collaborative manufacturing. *Robotics and Computer-Integrated Manufacturing*, 2023. doi: 10.1016/j.rcim.2023.102608. URL <https://doi.org/10.1016/j.rcim.2023.102608>.
- Valentina De Simone; Valentina Di Pasquale; Valeria Giubileo; Salvatore Miranda. Human-robot collaboration: an analysis of workers performance. *Procedia Computer Science*, 2022. doi: 10.1016/j.procs.2022.01.355. URL <https://doi.org/10.1016/j.procs.2022.01.355>.
- Yizhi Liu; Mahmoud Habibnezhad; Houtan Jebelli; Vishal Monga. Worker-in-the-loop cyber-physical system for safe human-robot collaboration in construction. *Computing in Civil Engineering 2021*, 2022. doi: 10.1061/9780784483893.132. URL <https://doi.org/10.1061/9780784483893.132>.
- Benjamin Birch; C E M Griffiths; Adam Milton Morgan. Environmental effects on reliability and accuracy of mfcc based voice recognition for industrial human-robot-interaction. *Proceedings of the Institution of Mechanical Engineers Part B Journal of Engineering Manufacture*, 2021. doi: 10.1177/09544054211014492. URL <https://doi.org/10.1177/09544054211014492>.
- R. Raffik; Rithish Ramamoorthy Sathya; V Vaishali; S. Balavedhaa; Jyothi Lakshmi N. Industry 5.0: Enhancing human-robot collaboration through collaborative robots a review. 2023. doi: 10.1109/icaeca56562.2023.10201120. URL <https://doi.org/10.1109/icaeca56562.2023.10201120>.
- Pai Zheng; Shufei Li; Liqiao Xia; Lihui Wang; Aydin Nassehi. A visual reasoning-based approach for mutual-cognitive human-robot collaboration. *CIRP Annals*, 2022. doi: 10.1016/j.cirp.2022.04.016. URL <https://doi.org/10.1016/j.cirp.2022.04.016>.
- Marco Costanzo; Giuseppe De Maria; Gaetano Lettera; Ciro Natale. A multimodal approach to human safety in collaborative robotic workcells. *IEEE Transactions on Automation Science and Engineering*, 2021. doi: 10.1109/tase.2020.3043286. URL <https://doi.org/10.1109/tase.2020.3043286>.
- Yaxu Niu; Frederik Schulte; Rudy R. Negenborn. Human aspects in collaborative order picking letting robotic agents learn about human discomfort. *Procedia Computer Science*, 2021. doi: 10.1016/j.procs.2021.01.338. URL <https://doi.org/10.1016/j.procs.2021.01.338>.
- Guy Hoffman; Tapomayukh Bhattacharjee; Stefanos Nikolaidis. Inferring human intent and predicting human action in humanrobot collaboration. *Annual Review of Control, Robotics, and Autonomous Systems*, 2024. doi: 10.1146/annurev-control-071223-105834. URL <https://doi.org/10.1146/annurev-control-071223-105834>.
- Dylan P. Losey; Craig G. McDonald; Edoardo Battaglia; Marcia K. OMalley. A review of intent detection, arbitration, and communication aspects of shared control for physical human-robot interaction. *Applied Mechanics Reviews*, 2018. doi: 10.1115/1.4039145. URL <https://doi.org/10.1115/1.4039145>.
- Simone Luca Pizzagalli; Vladimir Kuts; Tauno Otto. User-centered design for human-robot collaboration systems. *IOP Conference Series Materials Science and Engineering*, 2021. doi: 10.1088/1757-899x/1140/1/012011. URL <https://doi.org/10.1088/1757-899x/1140/1/012011>.

- Zeynel Abidin Cil; Zixiang Li; Suleyman Mete; Eren Ozceylan. Mathematical model and bee algorithms for mixed-model assembly line balancing problem with physical humanrobot collaboration. *Applied Soft Computing*, 2020. doi: 10.1016/j.asoc.2020.106394. URL <https://doi.org/10.1016/j.asoc.2020.106394>.
- Andrea Sanna; Federico Manuri; Jacopo Fiorenza; Francesco De Pace. Bari: An affordable brain-augmented reality interface to support humanrobot collaboration in assembly tasks. *Information*, 2022. doi: 10.3390/info13100460. URL <https://doi.org/10.3390/info13100460>.
- Yanjun Shi; Weiming Shen; Lihui Wang; Francesco Longo; Letizia Nicoletti; Antonio Padovano. A cognitive digital twins framework for human-robot collaboration. *Procedia Computer Science*, 2022. doi: 10.1016/j.procs.2022.01.387. URL <https://doi.org/10.1016/j.procs.2022.01.387>.
- Sarah K. Hopko; Riya Khurana; Ranjana K. Mehta; Prabhakar R. Pagilla. Effect of cognitive fatigue, operator sex, and robot assistance on task performance metrics, workload, and situation awareness in human-robot collaboration. *IEEE Robotics and Automation Letters*, 2021. doi: 10.1109/lra.2021.3062787. URL <https://doi.org/10.1109/lra.2021.3062787>.
- Vincent Gbouna Zakka; Gaoyang Pang; Geng Yang; Zeyang Hou; Honghao Lyu; Zhangwei Yu; Zhibo Pang. User-interactive robot skin with large-area scalability for safer and natural human-robot collaboration in future telehealthcare. *IEEE Journal of Biomedical and Health Informatics*, 2021. doi: 10.1109/jbhi.2021.3082563. URL <https://doi.org/10.1109/jbhi.2021.3082563>.
- Wenzheng Heng; Geng Yang; Gaoyang Pang; Zhiqiu Ye; Honghao Lyu; Juan Du; Guodong Zhao; Zhibo Pang. Fluid-driven soft coboskin for safer humanrobot collaboration: Fabrication and adaptation. *Advanced Intelligent Systems*, 2020. doi: 10.1002/aisy.202000038. URL <https://doi.org/10.1002/aisy.202000038>.
- Aswin K Ramasubramanian; Robins Mathew; Matthew Kelly; Vincent Hargaden; Nikolaos Papakostas. Digital twin for humanrobot collaboration in manufacturing: Review and outlook. *Applied Sciences*, 2022. doi: 10.3390/app12104811. URL <https://doi.org/10.3390/app12104811>.
- Kyle B. Reed; Michael A. Peshkin. Physical collaboration of human-human and human-robot teams. *IEEE Transactions on Haptics*, 2008. doi: 10.1109/toh.2008.13. URL <https://doi.org/10.1109/toh.2008.13>.
- Nicky Mol; J. Micah Prendergast; David A. Abbink; Luka Peternel. Fitts' list revisited: An empirical study on function allocation in a two-agent physical human-robot collaborative position/force task. *arXiv*, 2025. doi: 10.1109/lra.2025.3632607. URL <http://arxiv.org/abs/2505.04722v2>.
- Tilen Brecelj; Tadej Petric. Stable heteroclinic channel networks for physical humanhumanoid robot collaboration. *Sensors*, 2023. doi: 10.3390/s23031396. URL <https://doi.org/10.3390/s23031396>.
- Bin Liu; Wenjun Xu; Jiayi Liu; Bitao Yao; Zude Zhou; Duc Truong Pham. Human-robot collaboration for disassembly line balancing problem in remanufacturing. 2019a. doi: 10.1115/msec2019-2919. URL <https://doi.org/10.1115/msec2019-2919>.
- Huiping Cheng; Wenjun Xu; Qingsong Ai; Quan Liu; Zude Zhou; Duc Truong Pham. Manufacturing capability assessment for human-robot collaborative disassembly based on multi-data fusion. *Procedia Manufacturing*, 2017. doi: 10.1016/j.promfg.2017.07.008. URL <https://doi.org/10.1016/j.promfg.2017.07.008>.
- Quan Liu; Zhihao Liu; Wenjun Xu; Quan Tang; Zude Zhou; Duc Truong Pham. Human-robot collaboration in disassembly for sustainable manufacturing. *International Journal of Production Research*, 2019b. doi: 10.1080/00207543.2019.1578906. URL <https://doi.org/10.1080/00207543.2019.1578906>.

- Gregory Lemasurier; Gal Bejerano; Victoria Albanese; Jenna Parrillo; Holly A. Yanco; Nicholas Amerson; Rebecca Hetrick; Elizabeth Phillips. Methods for expressing robot intent for human-robot collaboration in shared workspaces. *ACM Transactions on Human-Robot Interaction*, 2021. doi: 10.1145/3472223. URL <https://doi.org/10.1145/3472223>.
- Alexandre Angleraud; Amir Mehman Sefat; Metodi Netzev; Roel Pieters. Coordinating shared tasks in human-robot collaboration by commands. *Frontiers in Robotics and AI*, 2021. doi: 10.3389/frobt.2021.734548. URL <https://doi.org/10.3389/frobt.2021.734548>.
- Loris Roveda; Andrea Testa; Asad Ali Shahid; Francesco Braghin; Dario Piga. Q-learning-based model predictive variable impedance control for physical human-robot collaboration. *Artificial Intelligence*, 2022. doi: 10.1016/j.artint.2022.103771. URL <https://doi.org/10.1016/j.artint.2022.103771>.
- Loris Roveda; Andrea Testa; Asad Ali Shahid; Francesco Braghin; Dario Piga. Q-learning-based model predictive variable impedance control for physical human-robot collaboration (extended abstract). *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, 2023a. doi: 10.24963/ijcai.2023/790. URL <https://doi.org/10.24963/ijcai.2023/790>.
- Loris Roveda; Lorenzo Mantovani; Marco Maccarini; Francesco Braghin; Dario Piga. Optimal physical humanrobot collaborative controller with user-centric tuning. *Control Engineering Practice*, 2023b. doi: 10.1016/j.conengprac.2023.105621. URL <https://doi.org/10.1016/j.conengprac.2023.105621>.
- Nicole Berx; Wilm Decre; Joris De Schutter; Liliane Pintelon. A harmonious synergy between robotic performance and well-being in human-robot collaboration: A vision and key recommendations. *Annual Reviews in Control*, 2024. doi: 10.1016/j.arcontrol.2024.100984. URL <https://doi.org/10.1016/j.arcontrol.2024.100984>.
- Enhao Zheng; Yuhua Li; Zhiyu Zhao; Qining Wang; Hong Qiao. An electrical impedance tomography based interface for humanrobot collaboration. *IEEE/ASME Transactions on Mechatronics*, 2020. doi: 10.1109/tmech.2020.3039017. URL <https://doi.org/10.1109/tmech.2020.3039017>.
- Nicolas Rogeau; Aryan Rezaei Rad. Collaborative timber joint assembly: Augmented reality for multi-level human-robot interaction. *International Journal of Architectural Computing*, 2024. doi: 10.1177/14780771241286605. URL <https://doi.org/10.1177/14780771241286605>.
- Anna Konstant; Nitzan Orr; Michael Hagenow; Isabelle Gundrum; Yu Hen Hu; Bilge Mutlu; Michael Zinn; Michael Gleicher; Robert G. Radwin. Humanrobot collaboration with a corrective shared controlled robot in a sanding task. *Human Factors The Journal of the Human Factors and Ergonomics Society*, 2024. doi: 10.1177/00187208241272066. URL <https://doi.org/10.1177/00187208241272066>.
- Li Liu; Andrew Schoen; Curt Henrichs; Jingshan Li; Bilge Mutlu; Yajun Zhang; Robert G. Radwin. Human robot collaboration for enhancing work activities. *Human Factors The Journal of the Human Factors and Ergonomics Society*, 2022. doi: 10.1177/00187208221077722. URL <https://doi.org/10.1177/00187208221077722>.
- S. M. Mizanoor Rahman. Cyber-physical-social system between a humanoid robot and a virtual human through a shared platform for adaptive agent ecology. *IEEE/CAA Journal of Automatica Sinica*, 2017. doi: 10.1109/jas.2017.7510760. URL <https://doi.org/10.1109/jas.2017.7510760>.
- S. M. Mizanoor Rahman. Cognitive cyber-physical system (c-cps) for human-robot collaborative manufacturing. 2019. doi: 10.1109/sysose.2019.8753835. URL <https://doi.org/10.1109/sysose.2019.8753835>.
- Basel Alhaji; Michael Prilla; Andreas Rausch. Trust dynamics and verbal assurances in human robot physical collaboration. *Frontiers in Artificial Intelligence*, 2021. doi: 10.3389/frai.2021.703504. URL <https://doi.org/10.3389/frai.2021.703504>.

- Ashwin P. Dani; Iman Salehi; Ghananeel Rotithor; Daniel Trombetta; Harish Ravichandar. Human-in-the-loop robot control for human-robot collaboration: Human intention estimation and safe trajectory tracking control for collaborative tasks. *IEEE Control Systems*, 2020. doi: 10.1109/mcs.2020.3019725. URL <https://doi.org/10.1109/mcs.2020.3019725>.
- Arturo Daniel Sosa-Ceron; Hugo Gustavo Gonzalez-Hernandez; Jorge Antonio Reyes-Avendano. Learning from demonstrations in humanrobot collaborative scenarios: A survey. *Robotics*, 2022. doi: 10.3390/robotics11060126. URL <https://doi.org/10.3390/robotics11060126>.
- Costanza Messeri; Anna Bicchi; Andrea Maria Zanchettin; Paolo Rocco. A dynamic task allocation strategy to mitigate the human physical fatigue in collaborative robotics. *IEEE Robotics and Automation Letters*, 2022. doi: 10.1109/lra.2022.3143520. URL <https://doi.org/10.1109/lra.2022.3143520>.
- Matteo Meregalli Falerni; Vincenzo Pomponi; Hamid Reza Karimi; Matteo Lavit Nicora; Le Anh Dao; Matteo Malosio; Loris Roveda. A framework for humanrobot collaboration enhanced by preference learning and ergonomics. *Robotics and Computer-Integrated Manufacturing*, 2024. doi: 10.1016/j.rcim.2024.102781. URL <https://doi.org/10.1016/j.rcim.2024.102781>.
- Alexandra Weidemann; Nele Ruwinkel. The role of frustration in humanrobot interaction what is needed for a successful collaboration? *Frontiers in Psychology*, 2021. doi: 10.3389/fpsyg.2021.640186. URL <https://doi.org/10.3389/fpsyg.2021.640186>.
- Yee Yeng Liao; Kwangyeol Ryu. Task allocation in human-robot collaboration (hrc) based on task characteristics and agent capability for mold assembly. *Procedia Manufacturing*, 2020. doi: 10.1016/j.promfg.2020.10.026. URL <https://doi.org/10.1016/j.promfg.2020.10.026>.
- Elisa Prati; Valeria Villani; Fabio Grandi; Margherita Peruzzini; Lorenzo Sabatini. Use of interaction design methodologies for humanrobot collaboration in industrial scenarios. *IEEE Transactions on Automation Science and Engineering*, 2021. doi: 10.1109/tase.2021.3107583. URL <https://doi.org/10.1109/tase.2021.3107583>.
- Mohammad Anvaripour; Mahta Khoshnam; Carlo Menon; Mehrdad Saif. Fmg- and rnn-based estimation of motor intention of upper-limb motion in human-robot collaboration. *Frontiers in Robotics and AI*, 2020. doi: 10.3389/frobt.2020.573096. URL <https://doi.org/10.3389/frobt.2020.573096>.
- Dallin L. Cordon; Shaden Moss; Marc Killpack; John L. Salmon. Leveraging passive compliance of soft robotics for physical human-robot collaborative manipulation. *arXiv*, 2025. URL <http://arxiv.org/abs/2504.08184v1>.
- Zubair Iqbal; Maria Pozzi; Domenico Prattichizzo; Gionata Salvietti. Detachable robotic grippers for human-robot collaboration. *Frontiers in Robotics and AI*, 2021. doi: 10.3389/frobt.2021.644532. URL <https://doi.org/10.3389/frobt.2021.644532>.
- Muhammad Hamza Zafar; Even Falkenberg Langas; Filippo Sanfilippo. Exploring the synergies between collaborative robotics, digital twins, augmentation, and industry 5.0 for smart manufacturing: A state-of-the-art review. *Robotics and Computer-Integrated Manufacturing*, 2024. doi: 10.1016/j.rcim.2024.102769. URL <https://doi.org/10.1016/j.rcim.2024.102769>.
- Afonso Castro; Filipe Silva; Vitor Santos. Trends of human-robot collaboration in industry contexts: Handover, learning, and metrics. *Sensors*, 2021. doi: 10.3390/s21124113. URL <https://doi.org/10.3390/s21124113>.
- Alessandro Roncone; Olivier Mangin; Brian Scassellati. Transparent role assignment and task allocation in human robot collaboration. 2017. doi: 10.1109/icra.2017.7989122. URL <https://doi.org/10.1109/icra.2017.7989122>.
- Jake Brawer; Olivier Mangin; Alessandro Roncone; Sarah H. Widder; Brian Scassellati. Situated humanrobot collaboration: predicting intent from grounded natural language. 2018. doi: 10.1109/iro.2018.8593942. URL <https://doi.org/10.1109/iro.2018.8593942>.

- Olivier Mangin; Alessandro Roncone; Brian Scassellati. How to be helpful? supportive behaviors and personalization for human-robot collaboration. *Frontiers in Robotics and AI*, 2022. doi: 10.3389/frobt.2021.725780. URL <https://doi.org/10.3389/frobt.2021.725780>.
- Andrea Pupa; Cristian Secchi. A safety-aware architecture for task scheduling and execution for human-robot collaboration. *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021. doi: 10.1109/iros51168.2021.9636855. URL <https://doi.org/10.1109/iros51168.2021.9636855>.
- Federica Ferraguti; Renzo Villa; Chiara Talignani Landi; Andrea Maria Zanchettin; Paolo Rocco; Cristian Secchi. A unified architecture for physical and ergonomic humanrobot collaboration. *Robotica*, 2019. doi: 10.1017/s026357471900095x. URL <https://doi.org/10.1017/s026357471900095x>.
- Inaki Maurtua; Aitor Ibarguren; Johan Kildal; Loreto Susperregi; Basilio Sierra. Human-robot collaboration in industrial applications. *International Journal of Advanced Robotic Systems*, 2017. doi: 10.1177/1729881417716010. URL <https://doi.org/10.1177/1729881417716010>.
- Fabian Ranz; Vera Hummel; Wilfried Sihm. Capability-based task allocation in human-robot collaboration. *Procedia Manufacturing*, 2017. doi: 10.1016/j.promfg.2017.04.011. URL <https://doi.org/10.1016/j.promfg.2017.04.011>.
- Kyle Strabala; Min Kyung Lee; Anca Dragan; Jodi Forlizzi; Siddhartha S. Srinivasa. Learning the communication of intent prior to physical collaboration. *2012 IEEE RO-MAN: The 21st IEEE International Symposium on Robot and Human Interactive Communication*, 2012. doi: 10.1109/roman.2012.6343875. URL <https://doi.org/10.1109/roman.2012.6343875>.
- Stefanos Nikolaidis; David Hsu; Siddhartha S Srinivasa. Human-robot mutual adaptation in collaborative tasks: Models and experiments. *The International Journal of Robotics Research*, 2017. doi: 10.1177/0278364917690593. URL <https://doi.org/10.1177/0278364917690593>.
- Stefanos Nikolaidis; Minae Kwon; Jodi Forlizzi; Siddhartha Srinivasa. Planning with verbal communication for human-robot collaboration. *ACM Transactions on Human-Robot Interaction*, 2018. doi: 10.1145/3203305. URL <https://doi.org/10.1145/3203305>.
- Dario Antonelli; Dorota Stadnicka. Predicting and preventing mistakes in human-robot collaborative assembly. *IFAC-PapersOnLine*, 2019. doi: 10.1016/j.ifacol.2019.11.204. URL <https://doi.org/10.1016/j.ifacol.2019.11.204>.
- Niels Dehio; Joshua Smith; Dennis L. Wigand; Pouya Mohammadi; Michael Mistry; Jochen J. Steil. Enabling impedance-based physical humanmultirobot collaboration: Experiments with four torque-controlled manipulators. *The International Journal of Robotics Research*, 2022. doi: 10.1177/02783649211053650. URL <https://doi.org/10.1177/02783649211053650>.
- Marc Teyssier; Brice Parilusyan; Anne Roudaut; Jurgen Steimle. Human-like artificial skin sensor for physical human-robot interaction. 2021. doi: 10.1109/icra48506.2021.9561152. URL <https://doi.org/10.1109/icra48506.2021.9561152>.
- Pauline Maurice; Meghan E. Huber; Neville Hogan; Dagmar Sternad. Velocity-curvature patterns limit humanrobot physical interaction. *IEEE Robotics and Automation Letters*, 2017. doi: 10.1109/lra.2017.2737048. URL <https://doi.org/10.1109/lra.2017.2737048>.
- Kristin E. Schaefer; Ralph W. Brewer; Joe Putney; Edward Mottern; Jeffrey Barghout; Edward R. Straub. Relinquishing manual control: Collaboration requires the capability to understand robot intent. 2016. doi: 10.1109/cts.2016.0071. URL <https://doi.org/10.1109/cts.2016.0071>.
- Christopher Yee Wong; Lucas Vergez; Wael Suleiman. Vision- and tactile-based continuous multi-modal intention and attention recognition for safer physical humanrobot interaction. *IEEE Transactions on Automation Science and Engineering*, 2023. doi: 10.1109/tase.2023.3276856. URL <https://doi.org/10.1109/tase.2023.3276856>.

- Han Ding; Zhigang Wan; Yuansheng Zhou; Jinyuan Tang. A data-driven programming of the human-computer interactions for modeling a collaborative manufacturing system of hypoid gears by considering both geometric and physical performances. *Robotics and Computer-Integrated Manufacturing*, 2018. doi: 10.1016/j.rcim.2017.10.003. URL <https://doi.org/10.1016/j.rcim.2017.10.003>.
- Takanori Shibata; Toshihiro Tashima; K. Tanie. Emergence of emotional behavior through physical interaction between human and robot. 2003a. doi: 10.1109/robot.1999.774032. URL <https://doi.org/10.1109/robot.1999.774032>.
- Takanori Shibata; Toshihiro Tashima; K. Tanie. Subjective interpretation of emotional behavior through physical interaction between human and robot. 2003b. doi: 10.1109/icsmc.1999.825403. URL <https://doi.org/10.1109/icsmc.1999.825403>.
- Wei Win Loy; Jared Donovan; Markus Rittenbruch; Muge Belek Fialho Teixeira. Exploring ar-enabled humanrobot collaboration (hrc) system for exploratory collaborative assembly tasks. *Construction Robotics*, 2025. doi: 10.1007/s41693-025-00165-x. URL <https://doi.org/10.1007/s41693-025-00165-x>.
- Mai Lee Chang; Reymundo A. Gutierrez; Priyanka Khante; Elaine Schaertl Short; Andrea L. Thomaz. Effects of integrated intent recognition and communication on human-robot collaboration. 2018. doi: 10.1109/iros.2018.8593359. URL <https://doi.org/10.1109/iros.2018.8593359>.
- Leonel Rozo; Sylvain Calinon; Darwin G. Caldwell; Pablo Jimenez; Carme Torras. Learning physical collaborative robot behaviors from human demonstrations. *IEEE Transactions on Robotics*, 2016. doi: 10.1109/tro.2016.2540623. URL <https://doi.org/10.1109/tro.2016.2540623>.
- Mert Imre; Erhan Oztog; Yukie Nagai; Emre Ugur. Affordance-based altruistic robotic architecture for humanrobot collaboration. *Adaptive Behavior*, 2019. doi: 10.1177/1059712318824697. URL <https://doi.org/10.1177/1059712318824697>.
- Claudio Urrea. Fatigue-aware sub-second combinatorial auctions for dynamic cycle allocation in humanrobot collaborative assembly. *Mathematics*, 2025. doi: 10.3390/math13152429. URL <https://doi.org/10.3390/math13152429>.
- Maryam Rezayati; Grammatiki Zanni; Ying Zaoshi; Davide Scaramuzza; Hans Wernher van de Venn. Improving safety in physical human-robot collaboration via deep metric learning. *2022 IEEE 27th International Conference on Emerging Technologies and Factory Automation (ETFA)*, 2022. doi: 10.1109/etfa52439.2022.9921623. URL <https://doi.org/10.1109/etfa52439.2022.9921623>.
- Fatemeh Mohammadi Amin; Maryam Rezayati; Hans Wernher van de Venn; Hossein Karimpour. A mixed-perception approach for safe humanrobot collaboration in industrial automation. *Sensors*, 2020. doi: 10.3390/s20216347. URL <https://doi.org/10.3390/s20216347>.
- Linda van der Spaa; Jens Kober; Michael Gienger. Simultaneously learning intentions and preferences during physical human-robot cooperation. *Autonomous Robots*, 2024. doi: 10.1007/s10514-024-10167-3. URL <https://doi.org/10.1007/s10514-024-10167-3>.
- Wietse van Dijk; Saskia J. Baltrusch; Ezra Dessers; Michiel P. de Looze. The effect of human autonomy and robot work pace on perceived workload in human-robot collaborative assembly work. *Frontiers in Robotics and AI*, 2023. doi: 10.3389/frobt.2023.1244656. URL <https://doi.org/10.3389/frobt.2023.1244656>.
- Luca Gualtieri; Erwin Rauch; Renato Vidoni. Development and validation of guidelines for safety in human-robot collaborative assembly systems. *Computers & Industrial Engineering*, 2021a. doi: 10.1016/j.cie.2021.107801. URL <https://doi.org/10.1016/j.cie.2021.107801>.
- Luca Gualtieri; Erwin Rauch; Renato Vidoni. Methodology for the definition of the optimal assembly cycle and calculation of the optimized assembly cycle time in human-robot collaborative assembly. *The International Journal of Advanced Manufacturing Technology*, 2021b. doi: 10.1007/s00170-021-06653-y. URL <https://doi.org/10.1007/s00170-021-06653-y>.

- Alberto Romay; Stefan Kohlbrecher; Alexander Stumpf; Oskar von Stryk; Spyros Maniatopoulos; Hadas KressGazit; Philipp Schillinger; David C. Conner. Collaborative autonomy between highlevel behaviors and human operators for remote manipulation tasks using different humanoid robots. *Journal of Field Robotics*, 2017. doi: 10.1002/rob.21671. URL <https://doi.org/10.1002/rob.21671>.
- Behzad Sadrifaridpour; Yue Wang. Collaborative assembly in hybrid manufacturing cells: An integrated framework for humanrobot interaction. *IEEE Transactions on Automation Science and Engineering*, 2017a. doi: 10.1109/tase.2017.2748386. URL <https://doi.org/10.1109/tase.2017.2748386>.
- Chengxi Li; Pai Zheng; Peng Zhou; Yue Yin; Carman K.M. Lee; Lihui Wang. Unleashing mixed-reality capability in deep reinforcement learning-based robot motion generation towards safe humanrobot collaboration. *Journal of Manufacturing Systems*, 2024. doi: 10.1016/j.jmsy.2024.03.015. URL <https://doi.org/10.1016/j.jmsy.2024.03.015>.
- Hongyi Liu; Lihui Wang. Remote humanrobot collaboration: A cyberphysical system application for hazard manufacturing environment. *Journal of Manufacturing Systems*, 2019. doi: 10.1016/j.jmsy.2019.11.001. URL <https://doi.org/10.1016/j.jmsy.2019.11.001>.
- Jianpeng Chen; Sihan Huang; Xiaowen Wang; Pengfei Wang; Jiahao Zhu; Zhe Xu; Guoxin Wang; Yan Yan; Lihui Wang. Perception-decision-execution coordination mechanism driven dynamic autonomous collaboration method for human-like collaborative robot based on multimodal large language model. *Robotics and Computer-Integrated Manufacturing*, 2026a. doi: 10.1016/j.rcim.2025.103167. URL <https://doi.org/10.1016/j.rcim.2025.103167>.
- Omar Obidat; Jesse Parron; Rui Li; Julia Rodano; Weitian Wang. Development of a teaching-learning-prediction-collaboration model for human-robot collaborative tasks. *2023 IEEE 13th International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER)*, 2023a. doi: 10.1109/cyber59472.2023.10256441. URL <https://doi.org/10.1109/cyber59472.2023.10256441>.
- Pai Zheng; Shufei Li; Junming Fan; Chengxi Li; Lihui Wang. A collaborative intelligence-based approach for handling human-robot collaboration uncertainties. *CIRP Annals*, 2023b. doi: 10.1016/j.cirp.2023.04.057. URL <https://doi.org/10.1016/j.cirp.2023.04.057>.
- Patrik Gustavsson; Magnus Holm; Anna Syberfeldt; Lihui Wang. Human-robot collaboration towards new metrics for selection of communication technologies. *Procedia CIRP*, 2018. doi: 10.1016/j.procir.2018.03.156. URL <https://doi.org/10.1016/j.procir.2018.03.156>.
- Sichao Liu; Zhihao Liu; Qiang Qin; Xi Vincent Wang; Lihui Wang. Multimodal human-robot collaboration: advancements and future directions. *International Journal of Manufacturing Research*, 2025. doi: 10.1504/ijmr.2025.149872. URL <https://doi.org/10.1504/ijmr.2025.149872>.
- Xi Vincent Wang; Zsolt Kemeny; Jozsef Vancza; Lihui Wang. Humanrobot collaborative assembly in cyber-physical production: Classification framework and implementation. *CIRP Annals*, 2017b. doi: 10.1016/j.cirp.2017.04.101. URL <https://doi.org/10.1016/j.cirp.2017.04.101>.
- Yichen Wang; Jindan Feng; Jinshan Liu; Xiaojun Liu; Junfeng Wang. Digital twin-based design and operation of human-robot collaborative assembly. *IFAC-PapersOnLine*, 2022. doi: 10.1016/j.ifacol.2022.04.209. URL <https://doi.org/10.1016/j.ifacol.2022.04.209>.
- Zhenyu Pan; Weiming Wang. Force sensing control for physical humanrobot interaction: A transformer-based action chunking approach. *Machines*, 2026b. doi: 10.3390/machines14020249. URL <https://doi.org/10.3390/machines14020249>.
- Mahdi Haghshenas-Jaryani; Muthu B. J. Wijesundara. A quasi-static model for studying physical interaction between a soft robotic digit and a human finger. *Volume 5A: 42nd Mechanisms and Robotics Conference*, 2018. doi: 10.1115/detc2018-85629. URL <https://doi.org/10.1115/detc2018-85629>.

- Apostolos Kalatzis; S. M. Mizanoor Rahman; Vishnunarayan Girishan Prabhu; Laura Stanley; Mike P. Wittie. A multimodal approach to investigate the role of cognitive workload and user interfaces in human-robot collaboration. *INTERNATIONAL CONFERENCE ON MULTIMODAL INTERACTION*, 2023. doi: 10.1145/3577190.3614112. URL <https://doi.org/10.1145/3577190.3614112>.
- Gaoyang Pang; Jia Deng; Fangjinhua Wang; Junhui Zhang; Zhibo Pang; Geng Yang. Development of flexible robot skin for safe and natural humanrobot collaboration. *Micromachines*, 2018. doi: 10.3390/mi9110576. URL <https://doi.org/10.3390/mi9110576>.
- Hyomin Kim; Woosung Yang. Variable admittance control based on humanrobot collaboration observer using frequency analysis for sensitive and safe interaction. *Sensors*, 2021. doi: 10.3390/s21051899. URL <https://doi.org/10.3390/s21051899>.
- Jing You; Zhiyuan Wu; Wei Wei; Ning Li; Yuhua Yang. Evolution of industrial robots from the perspective of the metaverse: Integration of virtual and physical realities and humanrobot collaboration. *Applied Sciences*, 2024. doi: 10.3390/app14146369. URL <https://doi.org/10.3390/app14146369>.
- Uqba Othman; Erfu Yang. An overview of human-robot collaboration in smart manufacturing. *2022 27th International Conference on Automation and Computing (ICAC)*, 2022. doi: 10.1109/icac55051.2022.9911168. URL <https://doi.org/10.1109/icac55051.2022.9911168>.
- Berat Mert Albaba; Yldray Yldz. Modeling cyber-physical human systems via an interplay between reinforcement learning and game theory. *Annual Reviews in Control*, 2019. doi: 10.1016/j.arcontrol.2019.10.002. URL <https://doi.org/10.1016/j.arcontrol.2019.10.002>.
- Karin Bogner; Ulrich Pferschy; Roland Unterberger; Herwig Zeiner. Optimised scheduling in humanrobot collaboration a use case in the assembly of printed circuit boards. *International Journal of Production Research*, 2018. doi: 10.1080/00207543.2018.1470695. URL <https://doi.org/10.1080/00207543.2018.1470695>.
- Tadele Belay Tuli; Martin Manns; Sebastian Zeller. Human motion quality and accuracy measuring method for humanrobot physical interactions. *Intelligent Service Robotics*, 2022. doi: 10.1007/s11370-022-00432-8. URL <https://doi.org/10.1007/s11370-022-00432-8>.
- Jing Luo; Xiangyu Zhou; Yifan Zhu; Yu Li; Chaoyi Zhang; Keao Wang; Zhaohong Mai; Chao Zeng. A humanrobot collaboration control framework for supernumerary robotic limbs. *Journal of Field Robotics*, 2026. doi: 10.1002/rob.70025. URL <https://doi.org/10.1002/rob.70025>.
- Chengyun Wang; Tingting Zhang. Dynamic human-like motion planning method of humanrobot collaboration based on movement primitives and double dqn. *Industrial Robot: the international journal of robotics research and application*, 2025a. doi: 10.1108/ir-07-2025-0249. URL <https://doi.org/10.1108/ir-07-2025-0249>.
- Haotian Liu; Yuchuang Tong; Guanchen Liu; Zhaojie Ju; Zhengtao Zhang. Idagc: Adaptive generalized human-robot collaboration via human intent estimation and multimodal policy learning. *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025b. doi: 10.1109/iros60139.2025.11246003. URL <https://doi.org/10.1109/iros60139.2025.11246003>.
- Haotian Liu; Yuchuang Tong; Zhengtao Zhang. Dtrt: Enhancing human intent estimation and role allocation for physical human-robot collaboration. *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025c. doi: 10.1109/icra55743.2025.11127335. URL <https://doi.org/10.1109/icra55743.2025.11127335>.
- Xingyu Yang; Zhengxue Zhou; Leihui Li; Xuping Zhang. Collaborative robot dynamics with physical humanrobot interaction and parameter identification with pinn. *Mechanism and Machine Theory*, 2023. doi: 10.1016/j.mechmachtheory.2023.105439. URL <https://doi.org/10.1016/j.mechmachtheory.2023.105439>.

- Zhen Deng; Jinpeng Mi; Dong Han; Rui Huang; Xiaofeng Xiong; Jianwei Zhang. Hierarchical robot learning for physical collaboration between humans and robots. 2017. doi: 10.1109/robio.2017.8324507. URL <https://doi.org/10.1109/robio.2017.8324507>.
- Zhengxue Zhou; Xingyu Yang; Hao Wang; Xuping Zhang. Coupled dynamic modeling and experimental validation of a collaborative industrial mobile manipulator with human-robot interaction. *Mechanism and Machine Theory*, 2022. doi: 10.1016/j.mechmachtheory.2022.105025. URL <https://doi.org/10.1016/j.mechmachtheory.2022.105025>.
- Chao Zhang; Guanghui Zhou; Dongxu Ma; Rui Wang; Jiacheng Xiao; Dan Zhao. A deep learning-enabled human-cyber-physical fusion method towards human-robot collaborative assembly. *Robotics and Computer-Integrated Manufacturing*, 2023. doi: 10.1016/j.rcim.2023.102571. URL <https://doi.org/10.1016/j.rcim.2023.102571>.
- Jun Huang; Duc Truong Pham; Yongjing Wang; Ji Chunqian; Wenjun Xu; Quan Liu; Zude Zhou. A strategy for human-robot collaboration in taking products apart for remanufacture. *FME Transaction*, 2019. doi: 10.5937/fmet1904731h. URL <https://doi.org/10.5937/fmet1904731h>.
- Zhenrui Ji; Quan Liu; Wenjun Xu; Bitao Yao; Jiayi Liu; Zude Zhou. A closed-loop brain-computer interface with augmented reality feedback for industrial human-robot collaboration. doi: 10.21203/rs.3.rs-283263/v1. URL <https://doi.org/10.21203/rs.3.rs-283263/v1>.
- Heng Zhang; Sheng Qi; Jiawei Hu; Xinjun Sheng; Zhenhua Xiong; Xiangyang Zhu. Cooperative transportation with mobile manipulator: A capability map-based framework for physical human-robot collaboration. *IEEE/ASME Transactions on Mechatronics*, 2022. doi: 10.1109/tmech.2022.3155601. URL <https://doi.org/10.1109/tmech.2022.3155601>.
- Xiaofeng Gao; Luyao Yuan; Tianmin Shu; Hongjing Lu; Song-Chun Zhu. Show me what you can do: Capability calibration on reachable workspace for human-robot collaboration. *arXiv*, 2021. doi: 10.1109/lra.2022.3144779. URL <http://arxiv.org/abs/2103.04077v3>.
- Shouyan Chen; Xinqi Sun; Zhijia Zhao; Meng Xiao; Tao Zou. An online humanrobot collaborative grinding state recognition approach based on contact dynamics and lstm. *Frontiers in Neurobotics*, 2022. doi: 10.3389/fnbot.2022.971205. URL <https://doi.org/10.3389/fnbot.2022.971205>.